

Derivatives and Integrals

An Annotated Discourse

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Abstract

This is a sample of many of the things you can do with PreTeXt. Sometimes the math makes sense, sometimes it seems to be written in the first person, sort of like this Abstract.

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2010 Math Subject Classification. Primary 00-01, 00-02; Secondary 00A99.

1 Introduction

We consider definite integrals of functions $f(x)$. For example,

$$\int_0^2 \sin^2(x) dx.$$

This is also a demonstration of the capabilities of [PreTeXt](#).

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2 The Fundamental Theorem

There is a remarkable theorem:¹

¹And fortunately we do not need to try to write it in the margin!

Theorem 2.1 The Fundamental Theorem of Calculus. *If $f(x)$ is continuous, and the derivative of $F(x)$ is $f(x)$, then*

$$\int_a^b f(x) dx = F(b) - F(a)$$

Proof. Left to the reader.² ■

You will find almost nothing about all this in the article [2], nor in the book [J.3.1], since they belong in some other article, but we can cite them out-of-order for practice anyway.

When we are writing we do not always know what we want to cite, or just where subsequent material will end up. For example, we might want a citation to [provisional cross-reference: some textbook about the FTC] or we might want to reference a later [provisional cross-reference: chapter about DiffEq's, and an_underscore].

We can also embed “todo”s in the source by making an XML comment that begins with the four characters `todo`, and selectively display them, so you may not see the one here in the output you are looking at now. Or maybe you do see it?

Because a definite integral can be computed using an antiderivative, we have the following definition.

Definition 2.2 Suppose that $\frac{d}{dx}F(x) = f(x)$. Then the **indefinite integral** of $f(x)$ is $F(x)$ and is written as

$$\int f(x) dx = F(x).$$

◆

3 Computing Integrals with Sage ($\int$)

Sage can compute definite integrals. The output contains the approximate numerical value of the definite integral, followed by an upper bound of the error in the approximation.

```
numerical_integral(sin(x)^2, (0, 2))
```

```
(1.189200623826982, 1.320277913471315e-14)
```

Given the Fundamental Theorem, we would find the antiderivative useful.

```
integral(sin(x)^2, x)
```

```
1/2*x - 1/4*sin(2*x)
```

The same command can be used to employ the antiderivative in the application of the Fundamental Theorem. Notice that the answer is *exact* and any further manipulation is likely to be simply producing a numerical approximation.

```
integral(sin(x)^2, (x, 0, 2))
```

²This is a footnote in a <proof> that is inside a <theorem>.

```
-1/4*sin(4) + 1
```

There are integrals you really do not want to evaluate, or you do not want your reader to evaluate. A Sage cell can be configured for display purposes only—you can look but you cannot touch.

```
integral(e^(x^2), x)
```

You can give a Sage element a `doctest` attribute, whose value mirrors the optional hash tags used in Sage doctests. Possible values are `random`, `long time`, `not implemented`, `not tested`, `known bug`, `absolute`, `relative`, and `optional`. The values `absolute` and `relative` refer to floating-point tolerances for equality and require a second attribute `tolerance` to specify a floating point value. The value `optional` refers to the test requiring that an optional Sage package be present. The name of that package should be provided in the `package` attribute.

The next cell is marked in the source as `doctest="random"`, and so is specified as unpredictable and not tested. But there is some “sample” output which will appear in the L^AT_EX version (and always be the same).

```
random()
```

```
0.11736021338650582
```

While the next cell is `random`, the returned value will never be more than 0.01 away from 12, since the `random()` function stays between 0 and 1. So we provide 12.005 as the expected answer, but test with an absolute tolerance of $\epsilon = 0.006$.

```
12 + 0.01*random()
```

```
12.005
```

Sage has some functions which affect output, generally making mathematics look more like mathematics via L^AT_EX syntax. This is a simple test, and you should see the variable and superscript in italics, properly formatted as output when viewed within HTML output. We have provided expected output for doctesting, but it is sort of silly to have this as part of L^AT_EX output, even if it is instructive.

```
pretty_print(html("$a^2$"))
```

```
<script type="math/tex">a^2</script>
```

Sage, and by extension, the Sage Cell Server, can interpret several languages. The next example has code in the R language, a popular open source language for statistics. As an author, you add the attribute `language="r"` to your `sage` element. (The default language is Sage, so you do not need to indicate that repeatedly.) Note that a language like R likes to use a “less than” character, `<`, special character in XML. You need to escape it by writing `<`; as we have done in the source for this example. (See the discussion in Subsection ??.)

As a reader you learn that the “Evaluate” button for a pre-loaded Sage cell will indicate the language in use.

```
ruth <- c(22, 25, 34, 35, 41, 41, 46, 46, 46, 47, 49, 54,
         54, 59, 60)
```

```
bonds <- c(16, 25, 24, 19, 33, 25, 34, 46, 37, 33, 42, 40,
          37, 34, 49, 73, 46, 45, 45, 5, 26, 28)
dimaggio <- c(12, 14, 20, 21, 25, 29, 30, 30, 31, 32, 32,
             39, 46)
summary(ruth)
summary(bonds)
summary(dimaggio)
boxplot(ruth, bonds, dimaggio)
```

The Sage Cell Server supports the following languages: `sage`, `gap`, `gp`, `html`, `maxima`, `octave`, `python`, `r`, and `singular`.

Here is another R cell. Unfortunately, it seems Sage’s `doctest` facility cannot be used easily with code from other languages. In the source for this example, we have employed the XML escape sequence, `<`; several times (see Subsection ??).

```
age <- c(25, 30, 56)
gender <- c("male", "female", "male")
weight <- c(160, 110, 220)
mydata <- data.frame(age, gender, weight)
summary(mydata)
cor(mydata$age, mydata$weight)
mean(mydata$age)
sd(mydata$age)
plot(mydata$age, mydata$weight)
```

The Sage Cell server imports a few important R packages. As of 2022-06-04 these are `deSolve`, `ggplot2`, `pracma`, `survey`, `swirl`, and `tidyverse`. This next example uses the `ggplot` library for both a data set and the plotting capabilities. Note the initial use of the `library()` function. This is a modified version of the “Bubble plot” example at [Top 50 ggplot2 Visualizations —The Master List](#).

```
# load package and data
library(ggplot2)
data(mpg, package="ggplot2")

mpg_select <- mpg[mpg$manufacturer %in% c("audi", "ford",
      "honda", "hyundai"), ]

# Scatterplot
g <- ggplot(mpg_select, aes(displ, cty)) +
  labs(subtitle="mpg: Displacement vs City Mileage",
       title="Bubble chart")

g + geom_jitter(aes(col=manufacturer, size=hwy)) +
  geom_smooth(aes(col=manufacturer), method="lm", se=F)
```

Here is a blank Sage cell that you may use for practice and experimentation with the commands above. Note that this cell allows a choice of languages, and is not linked with any of the previous cells, so a reader would need to start fresh, or cut/paste definitions from other cells. On the other hand a `<sage>` element with no content will also create an empty Sage cell for the reader’s use, but now it will be specific to a particular language and linked to others of the same language. Run the R cell above that defines the variable `ruth` and then try typing `summary(ruth)` in the cell below. You can make Sage blocks which are of `type="invisible"`, which will never be shown to a reader, but which get

doctested. Why do this? If some code produces an error, and you hope it is fixed someday, use an invisible block to raise the error. Once fixed, the doctest will fail, and you can adjust your commentary to suit. There is such a block right now, *but* you will need to go to the source to see it. Our maximum width for text, designed for readability, suggests you should format your Sage code with a maximum of about 54 characters. On a mobile device, the number of displayed characters might be as low as 28 in portrait orientation, and again around 50 in landscape. You can use a variety of techniques to shorten long lines, such as using intermediate variables. Since Sage is just a huge Python library, you can use any of Python’s facilities for handling long lines. These include a continuation character (which I try to avoid using) or natural places where you can break long lines, such as between entries of a list. Also, if writing loops or functions, you may wish to have your indentation be only two characters wide (rather than, say, four).

Sage output can sometimes be quite long, though this has improved with some changes in Sage’s output routines. Output code should be included primarily for doctesting purposes, and in this use, you may break at almost whitespace character and the doctesting framework will adjust accordingly. You may wish to show sample output in a static format, like a PDF, so you can consider formatting your output to fit the width constraints of that medium. Or you may even adjust exactly what is output, to keep it from being too verbose. Sage doctesting also allows for a wild-card style syntax which allows you to skip over huge chunks of meaningless or unpredictable output, such as tracebacks on error messages.

This paragraph is just a placeholder. It has handful of index entries, all starting with the letters “gas”, taken from *Indexing for Editors and Authors: A Practical Guide to Understanding Indexes* by Leise, Mertes, and Badgett. The intent is to test letter-by-letter versus word-by-word sorting of index entries. We use a word-by-word order, resulting in:

1. gas
2. gas masks
3. gas production
4. gas works
5. gasoline
6. gastritis

Titled Sage Cells.

```
integral(sin(x)^2, x)
```

```
1/2*x - 1/4*sin(2*x)
```

You can place Sage cells inside of a `paragraphs` if you want to give them a title, but no numbers, etc..

4 An Interesting Corollary

Objectives: Fundamental Structures

This is an <objectives> element you are reading, and this is its introduction. This early section has really grown and tries to accomplish many things. Not all of them are listed here.

1. Display various “blocks”, fundamental units of the flow.
2. More.
3. Evermore.

This concludes the (incomplete) objectives for this section, so now we can carry-on as before.

This is a cross-reference to one of the objectives above, forced to use the `phrase-global` form of the text. It should describe the objective as belonging to the *section* (rather than the *objectives*), since objectives are one-per-subdivision and are numbered based upon the containing division: [Objective 1 of Section ??](#). For comparison this is the (forced) `type-global` cross-reference: [Objective 4.1](#).

The Fundamental Theorem comes in two flavors, where usually one is a corollary of the other.

4.1 Second Version of FTC

Corollary 4.1 (Leibniz, Newton [1]) *Suppose $f(x)$ is a continuous function. Then*

$$\frac{d}{dx} \int_a^x f(t) dt = f(x). \quad (4.1)$$

Proof. We simply take the indicated derivative, applying Theorem ?? at (??)

$$\begin{aligned} \frac{d}{dx} \int_a^x f(t) dt &= \frac{d}{dx} (F(x) - F(a)) \\ &= \frac{d}{dx} F(x) - \frac{d}{dx} F(a) \\ &= f(x) - 0 = f(x). \end{aligned} \quad (4.2) \quad (4.3)$$

■

Justification. A justification, which is one of the variants of a proof. ■

Alternate Proof. You can have multiple proofs, and they can have titles which replace the word “Proof” as a heading. Here we just exercise displayed math with no automatic numbering, and an elective number on the middle equation.

$$\begin{aligned} \frac{d}{dx} \int_a^x f(t) dt &= \frac{d}{dx} (F(x) - F(a)) \\ &= \frac{d}{dx} F(x) - \frac{d}{dx} F(a) \\ &= f(x) - 0 = f(x) \end{aligned} \quad (4.4)$$

■

The alternative version of the Fundamental Theorem (FTC) in (??) is a compact way to express the result.

For testing purposes, there is a simple bare Sage Cell here.

```
c = 832
c
```

Example 4.2 A Mysterious Derivative! So if we define a function with its variable employed as a limit of integration, like so

$$K(z) = \int_{345}^z x^4 \sin(x^2) dx$$

then we get the derivative of that function so easily it seems like a mystery,

$$\frac{d}{dz}K(z) = z^4 \sin(z^2).$$

That's it.

For testing purposes, there is a simple Sage Cell here, buried inside an example that should be a knowl (embedded in the page).

```
2+2
```

We test a Sage cell inside a knowl, which should set the value of a variable that will be available to subsequent cells within the knowl.

```
a = 6
a
```

```
b = a + 10
b
```

Even if you ran the cell at the top of this page, within this knowl the value of the variable c is not known, so the next cell will cause an error.

```
c + 400
```



The Sage cells on a page will “remember” results computed elsewhere on the page. If you rely on this feature, remind your readers to evaluate all the necessary cells and that they perhaps need to be evaluated in a certain order.

```
c/2
```

There are some Sage cells in the previous (knowled) <example>. The results there are restricted to the knowl. In other words, the scope of those cells is the knowl. So if you opened the example and executed the Sage cells there, or if you skipped the example entirely, the next cell should not “know” the values of those variables and will raise an error.

```
a + b
```

We cross-reference the example just prior, Example ??, to test the simple Sage cells that will now be part of a cross-reference knowl (an external file).

Claim 4.3 An Equivalent Claim. *This claim is an equivalence: it is true if and only if it is correct.*

Proof. Our purpose here is to show how you can structure a proof with cases, such as an equivalence structured with the arrows typically used to demonstrate the two “directions” involved in the proof, by using the `@direction` attribute on a `<case>` element.

($\Rightarrow$) Nulla non lectus suscipit, bibendum leo quis, dignissim justo. In urna turpis, tincidunt id elementum id, faucibus ac tellus.

($\Leftarrow$) Quisque auctor ligula turpis, ut aliquam urna consectetur hendrerit. Aenean porta dolor et justo facilisis feugiat in sed sapien. Nullam porta ex et commodo semper.

Case 3b: The inductive step. A case may also have a `title`, whose formatting and structure is entirely up to the author. This then becomes the text of a cross-reference, as well.

Why Not Try This? A `<case>` (or any other element with a default title) did not always handle title-ending punctuation correctly. So we try an example title with a question mark.

($\Rightarrow$) *Necessity.* If you like, you can have both indications.

Case. No direction, no title, then just a generic title. ■

Exciting Proof! We test here that punctuation at the end of the title of a proof is handled correctly. ■

Exact Proof. This proof should fill exactly three lines (as of defaults in place 2018-12-31) and so the tombstone/Halmos should be on a fourth line, and then *flush right*. xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx. ■

Claim 4.4 A List of Equivalent Statements. *The following are equivalent.*

- (i) *This statement is equivalent to all those below.*
- (ii) *This statement is equivalent to the statement above and to all those below.*
- (iii) *This statement is equivalent to the two statements above and to the statement below.*
- (iv) *This statement is equivalent to all those above.*

Proof. Our purpose here is to show how you can structure a proof with cases to address the circular logic required to prove the equivalence of a list of statements, by using the `@direction` attribute on a `<case>` element. You should order the list of statements in the order that you would like to prove “this statement implies the next.”

((i) $\Rightarrow$ (ii)) Here we would prove that the first statement implies the second.

((ii) $\Rightarrow$ (iii)) Here we would prove that the second statement implies the third.

((iii) $\Rightarrow$ (iv)) *The trickiest case.* This time we include a title to describe the nature of this case. But we would still need to prove that the third statement implies the last.

((iv) $\Rightarrow$ (i)) *Wrap-around.* Finally, we would complete the cycle of logic by proving that the last statement implies the first. ■

We can also use @direction set to cycle in a stand-alone proof of our TFAE claim. If we include a @ref on the <proof> that points to the original claim, then the formatting of the markers on the statement list will be honored in our cases.

Proof. (??) Once again we will prove that the four statements in ?? are equivalent.

((i) $\Rightarrow$ (ii)) Another argument that the first statement implies the second.

((ii) $\Rightarrow$ (iii)) And another argument that the second statement implies the third.

((iii) $\Rightarrow$ (iv)) *Not so tricky this time.* Why did we find it so difficult before to prove that the third statement implies the last?

((iv) $\Rightarrow$ (i)) *Wrap-around.* And once more we complete the cycle of logic. ■

A couple more times to check that the default list markers get applied to the directional cases properly.

Claim 4.5 Another List of Equivalent Statements. *The that creates the list below does not have @marker.*

1. *This statement is equivalent to the two below.*

2. *This statement is equivalent to both the statement above and to the statement below.*

3. *This statement is equivalent to the two above.*

Proof. You know the drill by now.

(1 $\Rightarrow$ 2) Does the first statement imply the second?

(2 $\Rightarrow$ 3) *The “middle” case.* Does the second statement imply the third?

(3 $\Rightarrow$ 1) And finally, does the third imply the first? ■

This proof includes a @ref to the preceding claim.

Proof. (??)

(1 $\Rightarrow$ 2) Does the first statement imply the second?

(2 $\Rightarrow$ 3) *The “middle” case.* Does the second statement imply the third?

(3 $\Rightarrow$ 1) And finally, does the third imply the first? ■

This proof does not include a @ref, and so the direction indicators get default markers.

Proof.

(1 $\Rightarrow$ 2) Does the first statement imply the second?

(2 $\Rightarrow$ 1) And finally, does the second imply the first? ■

4.2 A Pedagogical Note about Subsection ??

4.2.1 Symbolic and Numerical Integrals

The Fundamental Theorem explains why we use the same notation for a definite integral, which is a numerical calculation,¹ and an antiderivative, which is a

¹Which I think sometimes students lose sight of.

symbolic expression.

Checkpoint 4.6 Essay Question: Compare and Contrast. Write a short paragraph which compares, and contrasts, the definite and indefinite integral. This is an exercise which sits in the midst of the narrative, so is formatted more like an example or a remark. It can have a hint and a solution, but this one does not. It can have a title, which this one does.

Hint. Start writing!

4.2.2 Advice

Using an “integral sign” for an antiderivative (aka indefinite integral) would seem to make the Fundamental Theorem a *fait accompli*. So I would suggest not conflating the notation for two very different things until the Fundamental Theorem exposes them as being highly related.

Example 4.7 An Example of Structure. This is an example of an example with a bit more structure. Specifically, the example has a **title**, as usual, but then has a **statement**, which is separate from the **solution**. Why did we implement an example in two ways?

Solution. Authors asked for it and it seemed a very natural thing to do, even if we only had an unstructured version for a long time. ▲

Question 4.8 An Example of a Question. Any kind of question can be marked as such with <question>. Or similarly, as a <problem>. They behave identically to **examples**, such as the one preceding and are numbered along with theorems, examples. etc.

Solution 1. You can have a solution. Or several, even if you don’t ask a question.

Solution 2. See? ▲

Checkpoint 4.9 An Inline Exercise. There are lots of exercises in this sample article, but mostly they are in special exercise sections. Sometimes you just want to sprinkle some exercises through the narrative. We call these **inline exercises**, in contrast to **divisional exercises**. The inline exercises look a bit more like a theorem or definition, with titles and fully-qualified numbers.

These may also have hints, answers and solutions.

Hint. A good hint.

Answer. 42.

Solution. If your exercise feels like proving a theorem, then you might want to make some comments, but also clearly delineate which part of the solution is a the complete proof.

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin lorem diam, convallis in nulla sed, accumsan fermentum urna. Pellentesque aliquet leo elit, ut consequat nunc dapibus ac. Sed lobortis leo tincidunt, vulputate nunc at, ultricies leo. Vivamus purus diam, tristique laoreet purus eget, mollis gravida sapien. Nunc vulputate nisl ac mauris hendrerit cursus. Sed vel molestie velit. Suspendisse sem sem, elementum at vehicula id, volutpat ac mi. Nullam ullamcorper fringilla purus in accumsan. Mauris at nunc accumsan orci dictum vulputate id id augue. Suspendisse at dignissim elit, non euismod nunc. Aliquam faucibus magna ac molestie semper. Aliquam hendrerit sem sit amet metus congue tempor. Donec laoreet laoreet metus, id interdum purus mattis vulputate. Proin condimentum vitae erat varius mollis. Donec venenatis libero sed turpis pretium tempor.

Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra. Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros. ■

Example 4.10 An Example of with $\frac{1}{2}$ math formula $\int e^x dx$ in the title. Just for testing math in knows, and also extra whitespace in a <p>. ▲

There are many different blocks you can employ, and they mostly behave the same way. A <project> is very similar to a <question> or <problem>

Project 4.1 Start Exploring PreTeXt. You could grab the `minimal.xml` file from the `examples/minimal` directory and experiment with that.

Projects get their own independent numbering scheme, since they may be central to your textbook, workbook, or lab manual. If you process this sample article with level for project numbering set to 0 then you will get consecutive numbers from the beginning of your book, starting with 1.

Exploration 4.2 Exploring Explorations. This is an <exploration>. Other similar possibilities are <project>, <activity>, <task>, and <investigation>.

Note that projects, activities, explorations, tasks and investigations *share* the independent numbering scheme, so it is really only intended you use one of these. If you want a variant of the name (e.g. “Directed Activity”) you can use the <rename> facility (Subsection ??).

Solution. This is a “solution” to the exploration. In practice, you might choose to not make this visible for students, but instead include it as part of some guidance you might provide to instructors (e.g. an *Instructor’s Manual*).

This is quite the activity upcoming. This is a prelude authored within the activity element, but visually just prior.

Activity 4.3 Hints, Answers, Solutions. Another variant of these project-like items is to possibly include a <hint> and an <answer> before the <solution>.

Hint. Just a little help.

Answer. The result, but no help in getting there.

Solution. Everything to get it all done, in detail.

This was quite the activity just now. This is a postlude authored within the activity element, but visually just after.

Note 4.11 A Note on Remarks. <remark>, <convention>, <note>, <observation> and <warning> are designed to hold very simple contents, with no additional structure (no proofs, no solutions, etc.).

But they do carry a title and a number, can be the target of a cross-reference, and may be optionally knowlized in HTML with the `html.knowl.remark` processing switch.

And distinctly different from a <note> in a <biblio>².

An Aside with a *Formatted* Title. An <aside> is similar to a remark, but is not as critical to the narrative. It is not numbered, and so requires a title. It can be the target of a cross-reference. They are meant to be short, and so are not knowlized at their first appearance. If the content is appropriate,

²A gratuitous footnote to test prior bug confusing this with a <note> in a <biblio>.

these can be marked as <historical> or <biographical>, though longer items should use subdivisions (e.g. sections, subsections) instead.

An <exercise> can be structured with <task>.

Checkpoint 4.12 A very structured exercise. This is an over-arching introduction to the whole exercise. We follow with some tasks.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(a) *A super-simple task.*

This first task is very simple, just a paragraph. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(b) Now three paragraphs. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(c) *A title of a task that has a subtask with an <answer> for the Solutions.*

This second task is further divided by more tasks. This is its introduction. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(i) *A task with a title and an <answer> for the Solutions.*

A really simple subtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

A short paragraph, before an answer.

Answer. With a proof.

Proof. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis. ■

And a bit more to say.

(ii) A subtask with an answer. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Answer. Right! In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(iii) Three simple sub-sub-tasks. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(A) First subsubtask. Short paragraph.

(B) *A second three-deep subsubtask!*

Second subsubtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(C) Third subsubtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

The conclusion of the structured subtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(iv) A simple task as the last subtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

This concludes our structured second task. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(d) This third top-level task is intermediate in complexity, you are reading the **statement**, which is followed by more items. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Hint. One hint. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Answer 1. First answer. In interdum suscipit ullamcorper.

Answer 2. Second answer. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Solution. At last, the solution. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

This is a conclusion where you could summarize the exercise. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

The following <project> is nearly identical to the preceding <exercise>.

The next block is a project, demonstrating the use of the `task` element to structure its parts. You are reading the `prelude` now. The project has lots of nonsense words, so we can test spacing the nested items. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Project 4.4 A very structured project. This is an over-arching introduction to the whole project. We follow with some tasks. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(a) This first task is very simple, just a paragraph. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(b) Now three paragraphs. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(c) This second task is further divided by more tasks. This is its introduction.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

- (i) A really simple subtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

A short paragraph, before an answer.

Answer. With a proof.

Proof. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis. ■

And a bit more to say.

- (ii) A subtask with an answer. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Answer. Right! In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

- (iii) Two simple sub-sub-tasks. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(A) First subsubtask. Short paragraph.

(B) Second subsubtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

(C) Third subsubtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

The conclusion of the structured subtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

- (iv) A simple task as the last subtask. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

This concludes our structured second task. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

- (d) This third top-level task is intermediate in complexity, you are reading the **statement**, which is followed by more items. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Hint. One hint. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Answer 1. First answer. In interdum suscipit ullamcorper.

Answer 2. Second answer. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Solution. At last, the solution. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

This is a conclusion where you could summarize the project. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

This postlude appears visually outside the project, but is authored within, to make clear its attachment to the project. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis. The following **<example>**, from Elise Desgreniers, is structured with **<task>**.

Example 4.13 Notation mathématique 2. Écrivez chacun des exemples suivants avec les conditions pertinentes.

- (a) Soit l'ensemble $A = \{1, 2, 3, 4, 5\}$.

On constate que cet ensemble contient uniquement des entiers positifs allant de 1 à 5.

Donc, on peut écrire $A = \{x \mid x \in \mathbb{N} \text{ et } 1 \leq x \leq 5\}$.

(b) $\{3, 6, 9, 12, 15, \dots, 27, 30\}$

Hint. Ce sont des multiples de 3.

Answer. $\{x \mid x = 3y \text{ et } 1 \leq y \leq 10 \text{ et } y \in \mathbb{N}\}$

(c) $\{1, 3, 5, 7, 9, 11, \dots\}$

Hint. Ce sont des nombres impairs.

Answer. $\{x \mid x \bmod 2 = 1 \text{ et } x \in \mathbb{N}\}$

(d) $\{2, 3, 5, 7, 11, 13, 17, 19, 23, \dots\}$

Hint. Ce sont des nombres premiers.

Answer. $\{x \mid x \text{ est un nombre premier positif}\}$

(e) $\{1, 4, 9, 16, 25, 36, \dots, 961\}$

Hint. Ce sont des carrés parfaits.

Answer. $\{x \mid x = y^2 \text{ et } 1 \leq y \leq 31 \text{ et } y \in \mathbb{N}\}$

(f) $\{1, 8, 27, 64, 125, \dots\}$

Hint. Ce sont des cubes parfaits.

Answer. $\{x \mid x = y^3 \text{ et } y \in \mathbb{N}^*\}$



Notes or examples related to computation or technology can go in blocks of the same name.

Technology 4.14 Sample Use of Sage. This would be a good place to talk about Sage, including a cell or two.

```
diff(x^4, x)
```

```
4*x^3
```

But you might want to describe how to use some other calculator, or maybe some numerical method.

A <paragraphs> with a <project> with an <answer>. The solutions to a project (and similar) once did not migrate to the automatically-generated solutions.

Project 4.5 A simple project, no tasks, just an answer.

Answer. Here's the answer we are looking for.

4.2.3 Exercises

1. This is an exercise in an “Exercises” subdivision at the level of a subsubsection. There is no question other than if the numbering is appropriate. Here is a self-referential link: [Exercise 4.2.3.1](#).

The subsubsection has no title in the source, so one is provided automatically, and will adjust according to the language of the document.

Solution. This solution will migrate to a list of solutions in the back-matter. We include a `sidebyside` as a test.

This is a skinny paragraph which should be just 30% of the width.	And another skinny paragraph which should also be just 30% of the width.
--	---

2. An `<exercise>` can be structured with parts, called `<task>`. This is the `<introduction>`.
 - (a) Do this.
 - (b) And the other thing.

4.2.4 Reading Questions

A set of reading questions may have an `<introduction>`, perhaps for preparatory explanation.

If a student has logged in to the HTML version, then they can answer the reading questions directly in the book. Inline math LaTeX can be entered using `$....$` or `\(...\)` delimiters, and inline AsciiMath using backticks ``...`` as delimiters. Here are some ``gratuitous backticks`` to check that AsciiMath is only active in the answers to reading questions.

1. This is a **reading question** that you might have a student answer prior to a class session, based on reading part of the book. A quick glance before class can help you tailor class time to the specific needs of your students. The perfect reading question will reveal whether the student has read and understood the material, and will be difficult to answer if they have not. What do you think of that?
2. And a second one, with a cross-reference to the first, as a check on numbering: [Reading Question 4.2.4.1](#). Reading questions are allowed to have answers, but providing answers misses the point of a reading question, and the answer knowl interacts poorly with the mechanism used to allow students to answer directly in the book. Do you think the schema should ban answers to reading questions?

And for symmetry, a `<conclusion>`.

4.2.5 Glossary

A glossary may have a `<headnote>`, perhaps with some explanation. This glossary is a specialized division within a section. Placement in the back matter is another option, see the [Glossary](#).

bar. A part of **foobar**. See foobar.

foobar. A synonym for the acronym FUBAR.

4.2.6 Solutions for This Subsection

This is an introduction, where you might explain that this division of this subsection contains various hints, answers, solutions of inline exercises, divisional exercises, and/or project-like blocks. See the source to see just how this solutions division was built.

4.2.1 · Symbolic and Numerical Integrals

Checkpoint 4.6 Essay Question: Compare and Contrast.

Hint. Start writing!

4.2.2 · Advice

Checkpoint 4.9 An Inline Exercise.

Hint. A good hint.

Answer. 42.

Solution. If your exercise feels like proving a theorem, then you might want to make some comments, but also clearly delineate which part of the solution is a the complete proof.

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin lorem diam, convallis in nulla sed, accumsan fermentum urna. Pellentesque aliquet leo elit, ut consequat nunc dapibus ac. Sed lobortis leo tincidunt, vulputate nunc at, ultricies leo. Vivamus purus diam, tristique laoreet purus eget, mollis gravida sapien. Nunc vulputate nisl ac mauris hendrerit cursus. Sed vel molestie velit. Suspendisse sem sem, elementum at vehicula id, volutpat ac mi. Nullam ullamcorper fringilla purus in accumsan. Mauris at nunc accumsan orci dictum vulputate id id augue. Suspendisse at dignissim elit, non euismod nunc. Aliquam faucibus magna ac molestie semper. Aliquam hendrerit sem sit amet metus congue tempor. Donec laoreet laoreet metus, id interdum purus mattis vulputate. Proin condimentum vitae erat varius mollis. Donec venenatis libero sed turpis pretium tempor.

Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra. Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros. ■

Exploration 4.2 Exploring Explorations.

Solution. This is a “solution” to the exploration. In practice, you might choose to not make this visible for students, but instead include it as part of some guidance you might provide to instructors (e.g. an *Instructor’s Manual*).

Activity 4.3 Hints, Answers, Solutions.

Hint. Just a little help.

Answer. The result, but no help in getting there.

Solution. Everything to get it all done, in detail.

Checkpoint 4.12 A very structured exercise.

(c) A title of a task that has a subtask with an <answer> for the Solutions.

(i) A task with a title and an <answer> for the Solutions.

Answer. With a proof.

Proof. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis. ■

And a bit more to say.

(ii) **Answer.** Right! In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

- (d) **Hint.** One hint. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Answer 1. First answer. In interdum suscipit ullamcorper.

Answer 2. Second answer. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Solution. At last, the solution. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Project 4.4 A very structured project.

- (c) (i) **Answer.** With a proof.

Proof. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis. ■

And a bit more to say.

- (ii) **Answer.** Right! In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

- (d) **Hint.** One hint. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Answer 1. First answer. In interdum suscipit ullamcorper.

Answer 2. Second answer. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Solution. At last, the solution. In interdum suscipit ullamcorper. Morbi sit amet malesuada augue, id vestibulum magna. Nulla blandit dui metus, malesuada mollis sapien ullamcorper sit amet. Nulla at neque nisi. Integer vel porta felis.

Project 4.5 Answer. Here's the answer we are looking for.

4.2.3 · Exercises

4.2.3.1. Solution. This solution will migrate to a list of solutions in the backmatter. We include a `sidebyside` as a test.

This is a skinny paragraph which should be just 30% of the width.	And another skinny paragraph which should also be just 30% of the width.
--	---

And a conclusion to this solutions division, which may not be readily apparent as distinct from the final division's worth of solutions, but since it is not prefixed with a number, it may be different enough.

4.3 Theorem-Like Environments

There are a variety of pre-defined environments in PreTeXt. All take a title, and must have a statement. Some have proofs (theorems, corollaries, etc.), while some do not have proofs (conjectures, axioms, principles).

Principle 4.15 The Title Principle. *It is a fundamental principle that many elements can have a title. Try it and see. If you get better formatting, then it is being recognized. If it looks very plain, check the documentation and perhaps make a feature request.*

More precisely, `<theorem>`, `<corollary>`, `<lemma>`, `<algorithm>`, `<proposition>`, `<claim>`, `<fact>`, and `<identity>`, all behave exactly the same, requiring a statement (as a sequence of paragraphs) followed by an optional proof, and may have an optional title. The elements `<axiom>`, `<conjecture>`, `<principle>`, `<heuristic>`, `<hypothesis>`, and `<assumption>` are functionally the same, barring a proof (since they would never have one!). Definitions are an exception, as it is natural to place `<notation>` within—see the source for Definition ?? for an example.

4.4 Linking Sage Cells

Sage cells share their results on a per-webpage basis, or a per-knowl basis, so if you move to a new chapter, section, or subsection that happens to be on another webpage, your Sage computations are gone and you start fresh. But maybe you need some results from elsewhere. As an author, you can make an exact copy of a cell in another location by placing the code in an external file, which is pure text, freed from any need to format for XML processing. So, in particular, there is no need to escape ampersands and angle brackets, nor is there employment of the CDATA mechanism. But the real value is that there is just one version to edit, and any changes will be reflected in both copies. We demonstrate this in the sample book, since it has the `xinclude` mechanism in place. In the chapter on groups, find the section on Sage and then find the

discussion of subgroups, and you will find an example of two identical Sage cells produced from one source file.

You can also specify certain cells to be auto-evaluated, by setting the `@auto-evaluate` attribute to `yes`. The resulting cell will not have a button for evaluation (so editing it would be pointless). See the source of this sample article for the two examples below.

2023-08-17: support just now is for the use case of a small portion of code, not a huge library of helper routines.

Two cells with the default language `sage`.

```
k = 404
```

```
2*k
```

Two cells with language `python`.

```
k = 4112
print("Assigned an initial value to the variable k")
```

```
k = 2*k
print(k)
```

4.5 Hierarchy

Structure. This section of this article has subsections and subsubsections. In a book you can have chapters enclosing multiple sections. There is one finer subdivision, it is achieved with the `paragraphs` element.

It is basically a sequence of paragraphs, where the first one gets an inline title. You are reading the second, and final, paragraph of one right now. It is useful for organizing very short documents, where numbered subdivisions might be overkill.

A Second Paragraphs. This is a second consecutive `paragraphs` element, so should seem related to its title, but distinct from the two paragraphs in the grouping with the title “Structure” immediately prior.

Assemblages: Collections and Summaries.

An `<assemblage>` is a collection, or summary, that does not have much structure to it. So you are limited to paragraphs and friends (`p`, `blockquote`, `pre`) and side-by-sides that do not contain captioned items (`sidebyside`, `sbsgroup`). The intent is that contents are not numbered, so cannot be cross-referenced individually, and so also do not become knowls. You may place `<image>`, `<tabular>`, and `<program>` inside a `<sidebyside>`, in addition to other objects that do not have captions. Note that `p` may by extension contain lists (`ol`, `ul`, `dl`). Despite limited structure, the presentation should draw attention to it, because the contents should be seen as more important in some way. It should be “highlighted” in some manner. If you need to connect the entire assemblage with material elsewhere, you can do that with the usual `xref/xml:id` mechanism.

What have we seen so far in this (disorganized) sample?

- Theorems, definitions and corollaries. (Section ??)

- Sage cells, including with R. (Section ??)
- Lots of document structure, like introductions and conclusions (next). (Section ??)

A sample table, as a `tabular` inside a `sidebyside` with no caption, follows.

A	B	C
Uno	Dos	Tres

This is a small `assemblage` with no title, simply to make sure the surrounding box behaves properly, especially for $\text{\LaTeX}$ output.

Assemblages containing $\mu\forall\tau\mathbb{H} = \emptyset\kappa$.

It is acceptable for an assemblage to contain mathematical content, even in its title.

4.6 Introductions and Conclusions

An Introductory Introduction. Any subdivision may have a sequence of paragraphs within an `<introduction>` that precedes subsequent further subdivisions. You are reading one now. They are always leaves of the document structure, so are rendered on some pages that reference the following subdivisions.

An introduction or conclusion is an extremely restrictive container with simple presentation. A title is optional (and probably not advisable). Content is meant to be short and unstructured, in particular, nothing that can be numbered is allowed. If this feels *too* restrictive, then place your content in an initial numbered subdivision and perhaps title it “Introduction”. Or make your entire subdivision unstructured and place whatever you want into it.

This ends this introduction to introductions.

4.6.1 Test One

An intervening subsubsection just after an introduction.

4.6.2 Test Two

An intervening subsection section which contains an `<exercises>` division which must be at the level of a Subsubsubsection.

Checkpoint 4.16 An inline exercise to examine any clash with divisional exercises below.

Answer. An answer so there is something to appear in a `<solutions>`.

What Did You Learn?

1. A mock exercise to appease validation.

Answer. An answer so there is something to appear in a `<solutions>`.

2. And a second to help with formatting the division heading.

Exercises

1. A mock exercise to appease validation.

Answer. An answer so there is something to appear in a `<solutions>`.

2. And a second to help with formatting the division heading.

4.6.3 Test Three

An intervening subsubsection just before a conclusion.

Entirely analogous to introductions are conclusions. Any subdivision may have a sequence of paragraphs within a `<conclusion>` that follows previous further subdivisions. You are reading one now. They are always leaves of the document structure, so are rendered on some pages that reference the preceding subdivisions.

This concludes this conclusion (and this subsection and this section).

4.7 Some Paragraph-Level Markup

Text within a paragraph may be *emphasized* with `<em>` or if you want to take it to the next level you can identify the text as an *alert* with `<alert>`.

Similarly, within a paragraph, you can identify edits between versions as *inserted text that has been added* with `<insert>` or as *deleted text that has been removed* with `<delete>`. Note that these identified edits are slightly different than *stale text that you want to retain, but which is no longer relevant*, which is accomplished with `<stale>`. The original request for stale text came from an instructor with an online list of student topics for presentations, and as students claimed topics they were marked as no longer available for other students.

If you need a “fill-in blank”, like this _____, it can be obtained with an empty `<fillin>` element that defaults to roughly a 10-character width. You can use the `@characters` attribute to make the rule longer or shorter, such as a 40-character blank: _____. The character count is approximate, based on typical character widths within a proportional font carrying English language text. Adjust to suit, or request a language-specific adjustment if it is critical.

This paragraph is intended to make a `<fillin>` appear right at the start of _____ the second line in print and then the next paragraph has nothing but a `<fillin>`. Both are for testing purposes.

The following are `<fillin>` with `@rows` and/or `@cols` attributes (at least one of which is greater than 1): _____ (2×1 array), _____ (1×3 array), _____ (2×3 array).

Long after we started this mess, we added PreTeXt tags to mark up tags and attributes. The elements are: `<tag>`, `<tage>`, `<attr>`. Examples of how these render are (respectively): `<section>`, `<hash/>`, `@width`. Perhaps this document will make greater use of these tags.

We supply two provisional cross-references for testing purposes only: [provisional cross-reference: a first incomplete cross-reference], [provisional cross-reference: a second incomplete cross-reference].

A conclusion here, which we fill with some numbering tests.

This is a cross-reference to one of the outcomes, forced to use the `type-global` form of the text. It should describe the outcome as belonging to the *section* (rather than the *outcomes*), since outcomes are one-per-subdivision and are numbered based upon the containing division: Outcome 2 of Section ???. For comparison this is the (forced) `type-global` cross-reference: Outcome 4.2.

Outcomes: Fundamental Structures, Revisited

This is a `<outcomes>` element you are reading, and this is its introduction. This early section has really grown and we have tried to accomplish many things. Not all of them are listed here.

1. Display various “blocks”, fundamental units of the flow.
2. More, and this is what the cross-references above are pointing to.
3. Evermore.

This concludes the (incomplete) outcomes for this section, so now we can carry-on to the next section.

5 Some Facts and Figures

Because of the Fundamental Theorem¹, for every derivative we know, there is an antiderivative we might find useful. Because of the Fundamental Theorem of Calculus², we recycle the “ $\int$ ” symbol as notation for an antiderivative.

- Derivatives
 - (a) $\frac{d}{dx}x^n = nx^{n-1}$
 - (b) $\frac{d}{dx}e^x = e^x$
 - (c) $\frac{d}{dx}\cos(x) = -\sin(x)$
 - Antiderivatives
- i) $\int_{-1} x^n dx = \frac{x^{n+1}}{n+1}$ if $n \neq -1$

ii) $\int e^x dx = e^x$

iii) $\int \sin(x) dx = -\cos(x)$

Remark 5.1 You can³ gain a greater understanding of derivatives by studying the graphs of functions with their derivatives. Can⁴ you discern the derivative–antiderivative⁵ relationship in Figure ???

¹First test footnote

²Second test footnote

³Third test footnote

⁴Fourth test footnote

⁵Fifth test footnote

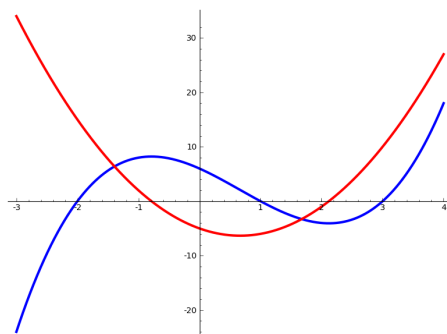


Figure 5.2 A function and its derivative

Lists⁶ can have multiple columns. With HTML items displayed in row-major order (horizontally first) and⁷ with L^AT_EX items are displayed in column-major order (vertically first). When one order, or the other, becomes workable in both variants, maybe we will be consistent in presentation. (Note that with just one row, it makes no difference.) We used it above for the two items—derivatives and integrals—where each item was a list of its own. Here are two more examples, one with short snippets and lots of columns, the other with lots of text in paragraphs.

- | | | | |
|----------|-----------|-----------|----------|
| 1. Red | 4. Purple | 7. Orange | 10. Aqua |
| 2. Blue | 5. Yellow | 8. Pink | 11. Cyan |
| 3. Green | 6. Black | 9. Salmon | 12. Puce |

- Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin lorem diam, convallis in nulla sed, accumsan fermentum urna. Pellentesque aliquet leo elit, ut consequat nunc dapibus ac. Sed lobortis leo tincidunt, vulputate nunc at, ultricies leo. Vivamus purus diam, tristique laoreet purus eget, mollis gravida sapien. Nunc vulputate nisl ac mauris hendrerit cursus. Sed vel molestie velit. Suspendisse sem sem, elementum at vehicula id, volutpat ac mi. Nullam ullamcorper fringilla purus in accumsan. Mauris at nunc accumsan orci dictum vulputate id id augue. Suspendisse at dignissim elit, non euismod nunc. Aliquam faucibus magna ac molestie semper. Aliquam hendrerit sem sit amet metus congue tempor. Donec laoreet laoreet metus, id interdum pu-

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⁶Sixth test footnote

⁷Seventh test footnote

tempus dui. Maecenas sollicitudin magna viverra, egestas velit nec, tristique sem. Cras iaculis mattis dui ac cursus. Integer volutpat, urna vel tempus convallis, erat nisi consectetur turpis, id varius dolor lorem vitae mauris. Phasellus erat orci, laoreet commodo gravida quis, congue in lacus. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Praesent at bibendum turpis. Pellentesque est nisl, dapibus at sagittis non, ultricies in nunc. Etiam ipsum arcu, porta sed feugiat eget, facilisis nec libero. Mauris tempor convallis felis.

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sed nisi nulla. Nam iaculis felis a enim blandit, at venenatis diam congue. Nulla augue diam, egestas eget fermentum nec, posuere eget risus. Praesent egestas nulla eros, eget accumsan augue euismod vel. Pellentesque pellentesque non erat vitae posuere. Curabitur lacus arcu, varius sed risus ut, ullamcorper tincidunt lorem. Sed et lacus dignissim, tincidunt nisl ac, porttitor sapien.

6 Some Advanced Ideas

The multi-row displayed mathematics in the proof of the Fundamental Theorem had equations aligned on the equals signs via the `&` character. Sometimes you don't want that. Here is an example with some differential equations, with each equation centered and unnumbered,

$$\begin{aligned}\mathcal{L}(y')(s) &= s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0) \\ \mathcal{L}(y'')(s) &= s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0).\end{aligned}$$

Just prior to this sentence, in the middle of this paragraph, is an `<idx>` and a `<notation>`, adjacent, but separated by some whitespace in the authored source. That insignificant whitespace will be removed always, which will be a (slightly) noticeable improvement in the L^AT_EX output. We test referencing notation here, placed *before* the sentence-ending period and right after some inline mathematics—for $\mathbb{Z}_n$.

L^AT_EX has a device where you can interrupt a sequence of equations with a small amount of text and preserve the equation alignment on either side. Here are two tests of that device, with aligned equations and non-aligned equations. Study the source to see use and differences. (The math does not make sense.)

Aligned and numbered first.

$$\mathcal{L}(y')(s) = s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0) \tag{6.1}$$

$$\mathcal{L}(y'')(s) = s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0). \tag{6.2}$$

And so it follows that,

$$\mathcal{L}(y')(s)^{++} = s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0) \tag{6.3}$$

$$\mathcal{L}(y'')(s)^5 = s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0). \tag{6.4}$$

Now with no numbers and no alignment. We include two cross-references in the `intertext` portion for testing.

$$\begin{aligned}\mathcal{L}(y')(s) &= s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0) \\ \mathcal{L}(y'')(s) &= s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0).\end{aligned}$$

First an external reference to [example.com](#) and internal cross-reference to Corollary `??`. And so it follows that,

$$\mathcal{L}(y')(s)^{++} = s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0)$$

$$\mathcal{L}(y'')(s)^5 = s^2 \mathcal{L}(y)(s) - sy(0) - y'(0) = s^2 Y(s) - sy(0) - y'(0).$$

Tables can get quite complex. Simple ones are simpler, such as this example of numerical computations for Euler’s method in just a bit.

But first we make a figure with two very simple tables next to each other. This causes the very first instance of `<table>` to actually be a “subtable”, which exposes a bug provoked by Emiliano Vega and fixed around 2020-08-06. (So we have to place this early to create the same behavior that exposed the bug.)

One	Two
(a) First	(b) Second

Figure 6.1 Buggy sub-tables

Table 6.2 Euler’s approximation for Duffing’s Equation with $h = 0.2$

i	t_i	x_i	y_i
0	0.00	0.0000	0.5000
1	0.20	0.1000	0.4800
2	0.40	0.1960	0.4560
3	0.60	0.2872	0.4295
4	0.80	0.3731	0.4027
5	1.00	0.4536	0.3783
6	1.20	0.5293	0.3591
7	1.40	0.6011	0.3480
8	1.60	0.6707	0.3474
9	1.80	0.7402	0.3603
10	2.00	0.8123	0.3900

7 Mathematics

To be able to create both $\text{\LaTeX}$ and HTML output (plus variations), we rely on MathJax, which in turn supports an extensive subset of the mathematical symbols normally available. The AMSMath symbol set is a good approximation. The PreTeXt Guide has a link to the complete list of macros supported by MathJax. We load the `AMSsymbols` library.

7.1 Basic Mathematics

The following is from the MathJax [demonstration page](#), an identity due to Ramanujan:

$$\frac{1}{\left(\sqrt{\phi\sqrt{5}}-\phi\right)e^{\frac{2}{5}\pi}}=1+\frac{e^{-2\pi}}{1+\frac{e^{-4\pi}}{1+\frac{e^{-6\pi}}{1+\frac{e^{-8\pi}}{1+\dots}}}}$$

And again, from the MathJax [demonstration page](#), Maxwell’s equations:

$$\begin{aligned}\nabla\times\vec{\mathbf{B}}-\frac{1}{c}\frac{\partial\vec{\mathbf{E}}}{\partial t}&= \frac{4\pi}{c}\vec{\mathbf{j}}\\ \nabla\cdot\vec{\mathbf{E}}&= 4\pi\rho\\ \nabla\times\vec{\mathbf{E}}+\frac{1}{c}\frac{\partial\vec{\mathbf{B}}}{\partial t}&= \vec{\mathbf{0}}\\ \nabla\cdot\vec{\mathbf{B}}&= 0\end{aligned}$$

Look back at the top of the source file of this document to see how to include your $\TeX$ macros just once. For best results keep your macros simple and semantic.

PreTeXt once provided modest built-in support for “slanted”, or “beveled”, or “nice” fractions. To wit, we mean fractions such as: $\frac{3}{8}$. Use the pre-defined `\sfrac{ }{ }` macro in your mathematics to achieve this presentation. The presentation in HTML is subpar, but could improve as MathJax provides support. It is now an author’s responsibility to add support for superior typesetting for PDF output by loading the `xfrac` $\LaTeX$ package with the following in `<docinfo>`:

```
<math-package latex-name="xfrac" mathjax-name="" />
```

which is what we have done here as a test. See the Guide for more details.

We consider a system of equations. We number the first and last equation (there are just two) and include an `@xml:id` attribute on each. We reference the whole system later as the range of equations from the first to the last.

$$\frac{dx}{dt} = x^2 - 4x - y + 4 \tag{7.1}$$

$$\frac{dy}{dt} = x^3 - y. \tag{7.2}$$

7.2 Displayed Mathematics

Display mathematics is achieved with the `<md>` tag (“math display”), always used within a paragraph (`<p>`). As a good example of how XML syntax is superior, you author n lines of equations by enclosing each line inside of a `<mrow>` tag, rather than using $n - 1$ separators (such as `\\`).

If you use no ampersands to express alignment (read ahead), then each equation is centered independently on the width of the text. This is implemented according to the AMSmath $\LaTeX$ package’s `gather` environment. Example:

$$\begin{aligned} \frac{dx}{dt} &= x^2 - 4x - y + 4 \\ \frac{dy}{dt} &= x^3 - y. \end{aligned}$$

An ampersand is used, in two ways, to describe positioning several equations per line, organized in columns. We have created the pre-defined $\LaTeX$ macro `\amp` as one way specify these, but the escape sequence `&` may be used also. The second, fourth, sixth, ... ampersands separate columns, and the spacing between columns will be provided automatically. The first, third, fifth, ... ampersands are alignment points for the equations in each column. Typically this is placed just prior to a binary operator, such as an equal sign (`\amp =`), or for a column of explanations or commentary, just prior to the `\text{ }` macro. Note that this scenario suggests always having an odd number of ampersands in each `<mrow>`. In the example below, alignment is on the equals sign in the first two columns, and provides left-justification to the explanations in the third column. N.B.: the use below of the `\text{ }` macro does not include mathematics within its argument. Doing so may yield unpredictable results depending on your choice of delimiters for the mathematics (and using an `<m>` tag will be ineffective).

$$\frac{dx}{dt} = x^2 - 4x - y + 4 \qquad \frac{dy}{dt} = x^3 - y \qquad x, y \text{ version}$$

$$\frac{dw}{dt} = z^3 - w \qquad \frac{dz}{dt} = z^2 - 4z - w + 4 \qquad z, w \text{ version}$$

PreTeXt will automatically detect the presence or absence of ampersands, but by defining macros for entire aligned equations, you can effectively hide the ampersands. So the `@alignment` attribute can override automatic detection. We use a simple L^AT_EX macro to demonstrate setting `alignment='align'` to override the use of a `gather` environment and use a `align` environment instead. Example:

$$\begin{aligned} \frac{dx}{dt} &= x^2 - 4x - y + 4 \\ \frac{dy}{dt} &= x^3 - y. \end{aligned}$$

The AMSmath L^AT_EX package's `alignat` environment is a third variant of alignment. It never happens automatically, you need to ask for it with `alignment="alignat"`. It is very similar to `align` but adds no space between the equation columns. So you can leave it that way, or you can add your own “extra” space to suit. Here is a previous example with no inter-column space:

$$\begin{aligned} \frac{dx}{dt} &= x^2 - 4x - y + 4 \frac{dy}{dt} = x^3 - y && x, y \text{ version} \\ \frac{dw}{dt} &= z^3 - w \qquad \frac{dz}{dt} = z^2 - 4z - w + 4z, w \text{ version.} \end{aligned}$$

This modified example has a middle row with three columns, while the other rows have just one column, as a test of our routines for determining the `<mrow>` with the greatest number of ampersands (and how many there are),

$$\begin{aligned} \frac{dw}{dt} &= z^3 - w \\ \frac{dx}{dt} &= x^2 - 4x - y + 4 \frac{dy}{dt} = x^3 - yx, y \text{ third column} \\ \frac{dw}{dt} &= z^3 - w. \end{aligned}$$

Final example demonstrates that ampersands in other objects (matrices here) can wreak havoc with computing the number of columns. So we provide yet another attribute to override automatic detection, `alignat-columns`. This is the number of *columns* not the number of *ampersands*. Generally, for c columns, there will be $2c - 1$ ampersands.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

One caveat: if your number of ampersands is even (see advice above about using an odd number) behavior should still be correct, as in next example.

If you want super-precise control over alignment of the terms of a system of equations (linear or not) you can use the `alignat` option to advantage by not including any extra space. This example is modified slightly from a post by Alex Jordan:

$$\begin{aligned} 2x + y + 3z &= 10 \\ x + z &= 6 \\ x + 3y + 2z &= 13. \end{aligned}$$

Beautiful.

Overlap Test. This is simply to test what happens when an aside bumps into something like the content below that may extend past the normal right content boundary. It likely only is of interest with the theme `default-modern`.

A long equation, to check layout on various screen sizes. This is Weil’s “explicit formula” for the Riemann ζ -function:

$$\sum_{\gamma} S_{-}(\gamma) = \frac{\log Q}{\pi} \hat{S}_{-}(0) + \frac{1}{2\pi} \sum_{j=1}^d \Re \left\{ \int_{-\infty}^{\infty} \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} + \mu_j \right) S_{-}(t) dt \right\} - \frac{d}{2\pi} \hat{S}_{-}(0) \log \pi. \quad (7.3)$$

Example 7.1 Excessive Display Mathematics. In print versions, a long run of displayed equations often needs to be broken across pages. If you are reading some other version of this, then there is nothing to see here. But for L^AT_EX output it could be interesting. First, with no extra effort, this page-long display should break naturally, no matter how the preceding material changes.

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$a^2 + b^2 = c^2$$

$$\alpha^2 + \beta^2 = \gamma^2$$

$$m^2 + n^2 = p^2$$

$$x^2 + y^2 = z^2$$

$$\begin{aligned}
a^2 + b^2 &= c^2 \\
\alpha^2 + \beta^2 &= \gamma^2 \\
m^2 + n^2 &= p^2 \\
x^2 + y^2 &= z^2 \\
a^2 + b^2 &= c^2 \\
\alpha^2 + \beta^2 &= \gamma^2 \\
m^2 + n^2 &= p^2 \\
x^2 + y^2 &= z^2 \\
a^2 + b^2 &= c^2 \\
\alpha^2 + \beta^2 &= \gamma^2 \\
m^2 + n^2 &= p^2 \\
x^2 + y^2 &= z^2 \\
a^2 + b^2 &= c^2 \\
\alpha^2 + \beta^2 &= \gamma^2 \\
m^2 + n^2 &= p^2 \\
x^2 + y^2 &= z^2 \\
a^2 + b^2 &= c^2 \\
\alpha^2 + \beta^2 &= \gamma^2 \\
m^2 + n^2 &= p^2.
\end{aligned}$$

In this version we have turned off page breaking for the entire display, but then allowed a break at every fourth equation, so you should see a reasonably attractive page break right after one of the $m^2 + n^2 = p^2$ equations.

$$x^2 + y^2 = z^2 \tag{7.4}$$

$$a^2 + b^2 = c^2 \tag{7.5}$$

$$\alpha^2 + \beta^2 = \gamma^2 \tag{7.6}$$

$$m^2 + n^2 = p^2 \tag{7.7}$$

$$x^2 + y^2 = z^2 \tag{7.8}$$

$$a^2 + b^2 = c^2 \tag{7.9}$$

$$\alpha^2 + \beta^2 = \gamma^2 \tag{7.10}$$

$$m^2 + n^2 = p^2 \tag{7.11}$$

$$x^2 + y^2 = z^2 \tag{7.12}$$

$$a^2 + b^2 = c^2 \tag{7.13}$$

$$\alpha^2 + \beta^2 = \gamma^2 \tag{7.14}$$

$$m^2 + n^2 = p^2 \tag{7.15}$$

$$x^2 + y^2 = z^2 \tag{7.16}$$

$$a^2 + b^2 = c^2 \tag{7.17}$$

$$\alpha^2 + \beta^2 = \gamma^2 \tag{7.18}$$

$$m^2 + n^2 = p^2 \tag{7.19}$$

$$\begin{aligned}
x^2 + y^2 &= z^2 & (7.20) \\
a^2 + b^2 &= c^2 & (7.21) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.22) \\
m^2 + n^2 &= p^2 & (7.23) \\
x^2 + y^2 &= z^2 & (7.24) \\
a^2 + b^2 &= c^2 & (7.25) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.26) \\
m^2 + n^2 &= p^2 & (7.27) \\
x^2 + y^2 &= z^2 & (7.28) \\
a^2 + b^2 &= c^2 & (7.29) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.30) \\
m^2 + n^2 &= p^2 & (7.31) \\
x^2 + y^2 &= z^2 & (7.32) \\
a^2 + b^2 &= c^2 & (7.33) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.34) \\
m^2 + n^2 &= p^2 & (7.35) \\
x^2 + y^2 &= z^2 & (7.36) \\
a^2 + b^2 &= c^2 & (7.37) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.38) \\
m^2 + n^2 &= p^2 & (7.39) \\
x^2 + y^2 &= z^2 & (7.40) \\
a^2 + b^2 &= c^2 & (7.41) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.42) \\
m^2 + n^2 &= p^2 & (7.43) \\
x^2 + y^2 &= z^2 & (7.44) \\
a^2 + b^2 &= c^2 & (7.45) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.46) \\
m^2 + n^2 &= p^2 & (7.47) \\
x^2 + y^2 &= z^2 & (7.48) \\
a^2 + b^2 &= c^2 & (7.49) \\
\alpha^2 + \beta^2 &= \gamma^2 & (7.50) \\
m^2 + n^2 &= p^2. & (7.51)
\end{aligned}$$

So. Do not take any extra steps and let L^AT_EX figure out the breaks. If you do not like a break, modify the <md> to go back to the AMSmath default behavior and not break at all. Ever. Or rather, go further and modify an individual <mrow> to suggest that it is a good place for a break. ▲

Displayed mathematics in L^AT_EX has a device called **intertext**, allowing for short snippets of text in-between displayed equations, but alignment continues across these short breaks. For any other output format besides PDF/L^AT_EX there really is not the technical means to make this continued alignment happen. So we mostly do the best we can. The following is a test example. Since almost all the equations are very similar in shape, it appears that there is alignment

throughout. However, for almost all output formats, the very last equation changes the horizontal positioning of the block with the final three equations, and they are not aligned with the previous equations. The exception is a PDF where we are able to take advantage of L^AT_EX's support and the equations should *all* be aligned on the equals signs.

$$A = Z + 1 \tag{7.52}$$

$$B = Y + 2 \tag{7.53}$$

One intertext.

$$C = X + 3 \tag{7.54}$$

$$D = W + 4 \tag{7.55}$$

$$E = V + 5 \tag{7.56}$$

Two intertext.

$$F = U + 6 \tag{7.57}$$

Three intertext.

$$G = T + 7 \tag{7.58}$$

$$H = S + 8 \tag{7.59}$$

$$I + J + K = P + Q + R + 9 + 10 + 11 \tag{7.60}$$

Now for some more testing of the <intertext> feature. Numbering some equations, and not others, requires some care in the conversion to L^AT_EX.

First, a numbered equation in the first group of equations.

$$A = B + C \tag{7.61}$$

Intertext: between first and second equations.

$$P = Q + R$$

Intertext: between second and third equations.

$$X = Y + Z$$

Second, a numbered equation in the intermediate group of equations.

$$A = B + C$$

Intertext: between first and second equations.

$$P = Q + R \tag{7.62}$$

Intertext: between second and third equations.

$$X = Y + Z$$

Third, a numbered equation in the final group of equations.

$$A = B + C$$

Intertext: between first and second equations.

$$P = Q + R$$

Intertext: between second and third equations.

$$X = Y + Z \tag{7.63}$$

Fourth, no numbered equations at all.

$$A = B + C$$

Intertext: between first and second equations.

$$P = Q + R$$

Intertext: between second and third equations.

$$X = Y + Z$$

This is a poorly-authored paragraph to test the conversion to HTML. There are two displayed equations, separated by a period ending the first one’s sentence, which should migrate into the display, and not leave behind an empty paragraph:

$$z + y = z.$$

$$a + b = c.$$

This final sentence should remain, inside another HTML paragraph, without the second equation’s period.

7.3 L^AT_EX Packages and MathJax Extensions

If you would like to enhance your mathematics by using a macro from a L^AT_EX package *and* there is a MathJax extension *which implements the same macro*, then you may use this with your mathematics as we demonstrate here.

This example is from Jason Underdown. The package is named `cancel` and is included in the TeXLive distribution, so is fairly standard. The particular macro being demonstrated is `\cancelto{}`.

$$\lim_{b \rightarrow \infty} \left[-\frac{1}{s} e^{\overset{0}{\cancel{sb}}} + \frac{1}{s} \right].$$

Look at the source of this article to see the package name being supplied in a `<math-package>` element within the `<docinfo>` section. That is the only setup required to make the macro usable in L^AT_EX and HTML output.

See the *PreTeXt Guide* subsection about “MathJax Extensions” for more detail.

7.4 Advanced Mathematics

MathJax is extremely capable in rendering a subset of L^AT_EX in web browsers, and improving all the time. You can get fairly fancy with some of its supported commands. In particular, if you need to mix in a few words with your mathematics, the `\text{}` macro is supported. For example, you might use an “if” or an “otherwise” in the definition of a piecewise function.

Consider that the first line below is text sandwiched in-between two Greek letters, wrapped in a `\text{}` macro. In HTML output we have taken care that the font for text material within display mathematics should match the font of the surrounding paragraph, as also happens with L^AT_EX output. The second line is nearly identical in the source, but is just naked text being rendered like a slew of variables.

$$\alpha \text{ is not equal to } \beta$$

$$\alpha is not equal to \beta$$

$$\alpha \neq \beta.$$

We are not suggesting here that using words in place of symbols, as in the first line, is a good practice. (It is not.)

The following example is a good stress-test of using the `\text{}` macro to achieve certain effects. Note the Unicode left and right smart quotes. This a contribution from Alex Jordan as part of his work on *APEX Calculus*.

$$y \rightarrow \frac{\sin(0)}{0} \rightarrow \text{“ } 0 \text{ ”}.$$

And another one from Alex. Note the use of the `\mathord{}` and `\mathrel{}` macros to control spacing around the mathematical symbols. Examine the source to see how the quotation marks have been authored with XML syntax for Unicode characters, since we do not allow most markup inside mathematics.

$$\zeta(1) = \sum_{n=1}^{\infty} \frac{1}{n} \text{ “=” } \prod_p \left(\frac{1}{1 - 1/p} \right) = \prod_p \left(\frac{1}{1 - p^{-1}} \right)$$

Generally, you cannot use any XML elements inside of the mathematics elements. An exception is the `<xref>` element which you might want to use to provide justifications for the steps of a derivation. Here is a visual example that is mathematically meaningless,

$A = B + C$	Corollary ??
$= D + E$	The Fundamental Theorem of Calculus
$= F + G$	A nice result.

Scott Beaver likes to write short chains of equalities all in one line, with the cross-references sitting on each equals sign. Here we test the L^AT_EX `\overset` and `\underset` macros wrapping a PreTeXt `<xref>`, with and without content, inside an `<md>` element. Note that `\stackrel` is obsolete, and `\overunderset` is not yet supported by MathJax (but see [GitHub #2704](#)). The mathematics is Scott’s, the reasons are totally unrelated to the math.

$$AC - AD \overset{??}{=} A(C - D) \underset{??}{=} A0_{n \times p} \overset{\text{Thm. } ??}{=} 0_{m \times p}$$

We suggest using cross-references that only display numbers (`<xref>` with `@text` set to `global`) since if you stick to elements like `<theorem>`, `<lemma>`, `<definition>`, or `<axiom>`, then the numbers will be unambiguous and the target of the cross-reference will contain full information. But note that if you mix in divisions, or perhaps figures, as reasons then there is a possibility that numbers will need to be qualified by their type. We have provided an abbreviation for one cross-reference to Theorem ?? (which will not benefit from automatic translation to other languages).

7.5 Numbered Equations

There is flexible control for numbering displayed equations, largely provided by the `@number` attribute on the `<md>` and `<mrow>` elements, taking values of `yes` or `no`. First, a single displayed equation, with a number:

$$x^2 + y^2 = z^2. \quad (7.64)$$

Now several displayed equations, all numbered.

$$x^3 + y^3 = z^3 \quad (7.65)$$

$$x^4 + y^4 = z^4 \quad (7.66)$$

$$x^5 + y^5 = z^5. \quad (7.67)$$

Similar, but now the first equation’s numbering is controlled by the global default value, the second is explicitly not numbered, and the third is explicitly numbered.

$$\begin{aligned} x^6 + y^6 &= z^6 \\ x^7 + y^7 &= z^7 \\ x^8 + y^8 &= z^8. \end{aligned} \quad (7.68)$$

The global default numbering choice is set in `<docinfo>`.

7.6 Local Tags on Equations

If you are not writing a research monograph, maybe (a) you will not use many numbered equations, or do not like the looks of them, or feel they scare your readers, and (b) maybe your cross-references are always local-ish, like strictly within an `<example>` or a `<proof>`. For this situation you can create, and employ, a “local” tag on a displayed equation. Nothing enforces the idea of what constitutes local, and there is nothing to stop you from using the same symbols more than once. With freedom comes responsibility.

Use the `@tag` attribute on an `<mrow>`, or directly on a single-line `<md>` (one with no `<mrow>` children). The value of the `@tag` attribute is a symbol name. The prefix `d` means “double”, and the prefix `t` means “triple”. So allowed values are

```
star, dstar, tstar
dagger, ddagger, tdagger
daggerdbl, ddaggerdbl, tdaggerdbl
hash, dhash, thash
maltese, dmaltese, tmaltese
```

Cross-references to these tagged equations happens in the usual way and should behave as expected. We test the double versions to make sure the symbols render properly in various output formats.

$$\begin{aligned} c^2 &= a^2 + b^2 & (**) \\ c^2 &= a^2 + b^2 & (\dagger\dagger) \\ c^2 &= a^2 + b^2 & (\ddagger\ddagger) \\ c^2 &= a^2 + b^2 & (\#\#) \\ c^2 &= a^2 + b^2 & (\text{\textcircled{X}}\text{\textcircled{X}}) \\ z^2 &= x^2 + y^2 & (7.69) \end{aligned}$$

Here are the local cross-references: (??), (??), (??), (??), (??). We test another farther away in Section ??, contrary to our advice above.

When there is nothing to align, the @tag attribute may live directly on a single-line <md>:

$$c^2 = a^2 + b^2. \tag{*}$$

This is equivalent to placing the content in a single <mrow> child carrying the @tag. A cross-reference to this equation looks like (??). Note that @tag and @number are mutually exclusive on a single-line <md>.

7.7 Commutative Diagrams

This diagram is authored by Tom Judson using the syntax of the AMS L^AT_EX “CD” package. Inside of a <md> element start with \begin{CD}. Remember to escape the less-than character.

$$\begin{array}{ccc} E[x]/\langle p(x) \rangle & \xrightarrow{\psi} & F[x]/\langle q(x) \rangle \\ \downarrow \sigma & & \downarrow \tau \\ E(\alpha) & \xrightarrow{\overline{\phi}} & F(\beta) \\ \downarrow & & \downarrow \\ E & \xrightarrow{\phi} & F \end{array}$$

While this package is not as flexible as some generic drawing packages, it has the advantage of full support by MathJax, and thus the HTML version will be more accessible.

7.8 Line-Breaking after Mathematics

As of 2021-05-14, in HTML output the next sentence should just fill a full line across the page. We take active measures to bind the concluding period to the final bit of mathematics, the variable *x*. The prevents a bad line break which could see the period *begin* a new line, all by itself. In the event that the line-breaking situation improves, we could relax these measures. This testing is only relevant to HTML output, not L^AT_EX output.

xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx xxx
xxx xxx xxx *x*.

7.9 Fonts and Mathematics

This section is about testing types and sizes of fonts, not so much about using different typefaces. First, one long displayed equation, which is designed to be full-width for L^AT_EX output when using defaults as of 2020-01-29 (commit defd4bffd462e7ea).

Start paragraph.

$$a^2+b^2+a^2+b^2+a^2+b^2+a^2+b^2+a^2+b^2+a^2+b^2+a^2+b^2+a^2+b^2+a^2+b^2$$

End paragraph.

The next paragraph has five ways to write the sine of *x*, bracketed by plain text versions. This tests font size and the fonts employed. The raw source of this paragraph is (spread over two lines)

```
sin x | <m>\sin x</m> | <m>\text{sin}\ x</m> | <m>\mathrm{sin}\ x</m> |
<m>\text{sin}x</m> | <m>sin x</m> | sin x
```

The five ways, from good to bad,

1. The best way, using `\sin`. Note the distance to the x .
2. With a `\text{}` macro.
3. With a `\mathrm{}` macro. Not recommended for PreTeXt.
4. Without a space. Note that the previous two require explicit spacing.
5. No extra effort, so $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ renders as a product of four variables.

`\sin x` | `\sin x` | `\sin x` | `\sin x` | `\sin x` | `\sin x` | `\sin x`

Finally a simple paragraph that places a text “x” next to a variable “x”.

`wordxxx + xxxword`

7.10 Special Characters in Text Mode

$\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ designates ten characters as **special** because they carry syntactic meaning. When these characters appear inside text-mode macros (such as `\text{}`) within mathematics, they must be escaped to produce their literal glyph. Seven of the ten have simple single-character escapes and are demonstrated in the following table. The remaining three (`~`, `^`, `\`) require macros from $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ ’s `textcomp` package (`\textasciitilde`, `\textasciicircum`, `\textbackslash`) which are not available in MathJax’s text mode as of 2026-04-03.

Character	Escaped form	Rendered
Hash #	<code>\#</code>	5#
Ampersand &	<code>\&</code>	P & Q
Percent %	<code>\%</code>	5%
Dollar \$	<code>\\$</code>	\$3
Underscore _	<code>_</code>	x_1
Left brace {	<code>\{</code>	{a, b}
Right brace }	<code>\}</code>	{a, b}

The hash character also requires escaping within the `\tag{}` macro. See (??) for an example where the `tag="dhash"` attribute produces a double hash as an equation tag.

7.11 Miscellaneous

In HTML output, a MathJax workaround for a Safari rendering bug was inserting extra spaces after textual subscripts and superscripts ([MathJax thread](#)). PreTeXt CSS now applies a correction. The following tests if the CSS fix is sufficient, and could be used to test the necessity of our fix in the future. Following is the original report, though NOT has been moved to a superscript:

$$T_{\text{clk}} - t_{\text{su}} > t_{\text{clk-Q}} + \max(t_{\text{XOR}}, t^{\text{NOT}}).$$

There should not be anything to see in $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ /PDF output. (2021-10-21)

7.12 Wide and Narrow Displays

This subsection stresses how a displayed equation is placed when it does, or does not, fit the width of the text, and how a number or a custom tag interacts with that placement. In print (PDF) output a display that fits is centered, with

any number at the right margin; a display too wide to fit is set flush with the left margin, so it overruns only the right margin and never the left.

A narrow display, numbered, centers with its number at the right margin:

$$a^2 + b^2 = c^2 \tag{7.70}$$

The same narrow display, unnumbered, simply centers:

$$a^2 + b^2 = c^2$$

A display wider than the text, numbered, is set flush left and overruns the right margin (and triggers an authoring warning):

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} = \sum_{i=1}^{15} x_i \tag{7.71}$$

The same wide display, unnumbered, behaves the same way:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} = \sum_{i=1}^{15} x_i$$

A wide display with a custom tag flushes left as well, the tag taking the place of a number:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} = \sum_{i=1}^{15} x_i \tag{*}$$

A multi-line display, numbered, with one narrow line and one too wide, exercises per-line numbering:

$$a + b = c \tag{7.72}$$

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} = \sum_{i=1}^{15} x_i \tag{7.73}$$

Inside a narrow container the available width is smaller, so a display centers within the container rather than the full text. Pushing a number all the way out to such a container's own right margin awaits a future refinement, so for now a number stays beside its centered equation. Here are two panels side by side, each with a numbered display.

A panel display:

$$a^2 + b^2 = c^2 \tag{7.74}$$

Another panel display:

$$u + v = w \tag{7.75}$$

A numbered display inside a block quotation, which is indented on both sides:

An indented, numbered display:

$$e^{i\pi} + 1 = 0 \tag{7.76}$$

A numbered display inside a doubly-nested list, where the item is indented twice:

1. Outer item.

(a) Inner item, with a numbered display:

$$\int_0^1 x^2 dx = \frac{1}{3} \tag{7.77}$$

8 Grouping Samples

While building or testing a rendering of PreTeXt, especially in HTML, it can be useful to see all the various elements that potentially create visual blocks in one place. So they are collected here.

We will omit content specific blocks like figures, images, tables, etc. as those elements have significant stress testing of their own elsewhere.

Please add any similar elements that are created or that you discover are missing from this page.

A title. There are more <aside>es below, but here is a solor one.

Objectives: Our goals

A minimal <objective>.

- Stress test HTML themes
- Locally, test objectives

8.1 Remark-like Blocks

Remark 8.1 A title. A minimal <remark>.

Convention 8.2 A title. A minimal <convention>.

Note 8.3 A title. A minimal <note>.

Observation 8.4 A title. A minimal <observation>.

Warning 8.5 A title. A minimal <warning>.

Insight 8.6 A title. A minimal <insight>.

8.2 Example-like Blocks

Example 8.7 A title. A minimal <example>.



Example 8.8 A title. A structured <example>.

(a) A structured <example>.

Hint. A <hint>

Answer. An <answer>

Solution. A <solution>

The <conclusion>.



Question 8.9 A title. A minimal <question>.



Problem 8.10 A title. A minimal <problem>.



Observation 8.11 A title. A minimal <observation>.

Warning 8.12 A title. A minimal <warning>.

Insight 8.13 A title. A minimal <insight>.

8.3 Theorem-like Blocks

Theorem 8.14 A title. A *minimal* <theorem>.

Theorem 8.15 A title. *A theorem with a proof.*

Proof. The proof of the theorem. ■

Corollary 8.16 A title. *A minimal <corollary>.*

Lemma 8.17 A title. *A minimal <lemma>.*

Algorithm 8.18 A title. *A minimal <algorithm>.*

Conundrum 8.19 A title. *A minimal <proposition>.*

Claim 8.20 A title. *A minimal <claim>.*

Fact 8.21 A title. *A minimal <fact>.*

Identity 8.22 A title. *A minimal <identity>.*

A title. A stand-alone proof. ■

8.4 Axiom-like Blocks

Axiom 8.23 A title. *A minimal <axiom>.*

Conjecture 8.24 A title. *A minimal <conjecture>.*

Principle 8.25 A title. *A minimal <principle>.*

Heuristic 8.26 A title. *A minimal <heuristic>.*

Hypothesis 8.27 A title. *A minimal <hypothesis>.*

Assumption 8.28 A title. *A minimal <assumption>.*

8.5 Definition-like Blocks

Definition 8.29 A title. A minimal <definition>. ◆

8.6 Aside-like Blocks

Three <aside>s are below.

A title. A minimal <aside>.

A title. A minimal <biographical>.

A title. A minimal <biographical>.

A title. A minimal <historical>.

8.7 Computation-like Blocks

Computation 8.30 A title. A minimal <computation>.

Technology 8.31 A title. A minimal <technology>.

Data 8.32 A title. A minimal <data>.

8.8 Project-like Blocks

Project 8.1 A title. A minimal <project>.

Activity 8.2 A title. A minimal <activity>.

Exploration 8.3 A title. A minimal <exploration>.

Investigation 8.4 A title. A minimal <investigation>.

8.9 Miscellaneous Blocks

A minimal <assemblage>.

A title. A final <aside> to test behavior towards the end of a page. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Outcomes: Our results

A minimal <outcome>.

- Stress test HTML themes
- Locally, test outcomes

Wrap-up. A minimal <conclusion>.

9 Entering Text in Paragraphs, Titles, Captions

XML, and therefore PreTeXt, is a markup language. But by and large, what you type into your source will be what you see in your output. So there is not much to say. Except that Subsection ?? eventually will be essential. However we do test various tricky situations here (which have technical explanations we avoid). See the Author’s Guide for a superior treatment of the topics addressed here.

9.1 Special XML characters

One of the goals of PreTeXt is to relieve an author of managing the numerous conflicts when mixing languages that use different characters for special purposes. But, of course, XML has its own special characters.

If you type a “less-than” symbol in your source, the XML processor thinks you are starting an opening, or closing, tag. So how do you get a less-than sign into your source so that it survives into your output, like this: < ? You use an **escaped version**. Type literally, the four characters < in your source. Then the XML processor will know you want the character and will not mistake it for a tag. But now we want to get an ampersand into our source like: &. How? Another escaped version of a character, literally the five characters &.

Otherwise, keys on your keyboard, even international versions, should be fine in your source and behave as expected. WYTIWYG = What You Type Is What You Get. So the principal concession to using XML markup is the following very simple rule.

Rather than pressing the < and & keys on your keyboard, instead always enter the escape sequences < and & as replacements.

Simple. And it will work in “running text,” verbatim text (like when authoring the content of <c> or <pre> elements), and mixed into L^AT_EX syntax to describe

mathematical expressions. XML has three other escape sequences `>`, `'`, and `"`, for the characters `>`, `'`, and `"` (respectively). But they seem largely unnecessary for authoring in PreTeXt, as we now demonstrate by typing them directly from our keyboard into our source: `>`, `'`, and `"`.

Checkpoint 9.1 How was `&` authored? Work it out, and then check the source here for the answer.

The next subsection should start on a new page in PDF output if the standard publisher file is used.

9.2 Quotations

The `<q>` tag will provide beginning and ending double quotations, while the `<sq>` tag will behave similarly but provide single quotes. Given the complexity of quotations, the different symbols used in different languages, and the over-simplified versions provided on keyboards, it is necessary to use markup.

“The roots of education are bitter, but the fruit is sweet.” (Aristotle)

‘It is always wise to look ahead, but difficult to look further than you can see.’ (Winston Churchill)

A large quote can be accommodated with the `<blockquote>` tag, which can carry within itself an `<attribution>` element.

The problem with writing a book in verse is, to be successful, it has to sound like you knocked it off on a rainy Friday afternoon. It has to sound easy. When you can do it, it helps tremendously because it’s a thing that forces kids to read on. You have this unconsummated feeling if you stop.

—Dr. Seuss

We say that again, to test a multiline attribution of a block quotation. Notice how the dash appears automatically, and that it is a **quotation dash** in HTML, distinct from other sorts of dashes.

The problem with writing a book in verse is, to be successful, it has to sound like you knocked it off on a rainy Friday afternoon. It has to sound easy. When you can do it, it helps tremendously because it’s a thing that forces kids to read on. You have this unconsummated feeling if you stop.

—Dr. Seuss
Children’s Author

Sometimes a quote may extend across several paragraphs. Or a balanced pair of quotations marks crosses an XML boundary, so we need left, right, single and double versions. (For example, see Section ?? on poetry.) Here are all four in a haphazard order: ”, ‘, “, ’. These should be a last resort, and *not* a replacement for the `q` and `sq` tags. The left/right versions are used for the following quote from Abraham Lincoln, which we have edited into two paragraphs.

“I am not bound to win, but I am bound to be true. I am not bound to succeed, but I am bound to live by the light that I have.

I must stand with anybody that stands right, and stand with him while he is right, and part with him when he goes wrong.”

And as a tests, we try some crazy combinations of quotes, which would normally give L^AT_EX some trouble where the quotation marks are adjacent. View these examples in PDF output, with lots of magnification and note the spacing between the quotation marks.

- “we use ‘single quotes inside of double quotes’”
- ““double quotes inside of single quotes” with more’
- “‘single quotes tight inside of double quotes’”
- ““double quotes tight inside of single quotes””
- An “““absurd test””” of two adjacent single quotes inside a pair of double quotes

- you would never do this, but a “pair of single quotes”

Quotation marks are different in different languages. The following paragraphs contain short examples.

English (United States). Primary only: “Government of the people, by the people, for the people, shall not perish from the earth.” Primary and secondary: “I have a dream that one day this nation will rise up and live out the true meaning of its creed: ‘We hold these truths to be self-evident, that all men are created equal.’” (Abraham Lincoln, Gettysburg Address; Martin Luther King Jr., I Have a Dream)

French (France). Primary only: « La liberté commence où l’ignorance finit. » Primary and secondary: « Elle se répétait : “J’ai un amant ! un amant !” se délectant à cette idée comme à celle d’une autre puberté qui lui serait survenue ». (Victor Hugo, *Les Misérables*; Gustave Flaubert, *Madame Bovary*)

French (Canada). Primary only: « La liberté commence où l’ignorance finit. » Primary and secondary: « Elle se répétait : ‘J’ai un amant ! un amant !’ se délectant à cette idée comme à celle d’une autre puberté qui lui serait survenue ». (Victor Hugo, *Les Misérables*; Gustave Flaubert, *Madame Bovary*)

German (Germany). Primary only: „Grau, teurer Freund, ist alle Theorie, und grün des Lebens goldner Baum.” Primary and secondary: „„Meine Luise’, sagte er mir, ‚hat mich zu Boden geworfen. Meine Luise wird mich auch aufrichten.’” (Johann Wolfgang von Goethe, *Faust I*; Friedrich Schiller, *Kabale und Liebe*)

9.3 Groupings

It is possible to make some other groupings like quotations, such as {some *emphasized text* grouped within braces}, or [a *Book Title* inside brackets], an “Article Title”, <some *foreign words* inside angle brackets>, or [just a bit of text within double brackets]. Some of these are used extensively by scholars who study texts to note various restorations or deletions. Note that the `<foreign>` element may have a `xml:lang` attribute.

Note that the angle brackets, `<` and `>`, are *not* the keyboard characters, `<` and `>`. Your best bet is to use the provided `<angles>` element when constructing a balanced pair. Similarly, `<dblbrackets>` is provided to make the double-bracket characters easily available, since they are likely not on your keyboard.

9.4 Characters, Symbols, and Constructions

Verbatim, inline text, is accomplished with the `<c>` element. One special consideration: if your editor splits a line in the middle, we will fix it. This sentence has two words that we purposely split with a “newline” character in the middle. In output you should see a space between the words.

Some keyboard characters are ambiguous. Is the character `'` an apostrophe or a right single quote? We presume the former, `'`, and provide markup as an alternative for the latter (described above). Is `/` used to separate words, or to form a fraction? We presume the former, `/`, and provide `<solidus/>`, `/`, for

the latter. We test some other characters straight from our US keyboard (with two being escape sequences).

~ ` ! @ # \$ % ^ & * () _ - + = [] { } | \ ; : ' " , < . > ? /

And again as verbatim text.

~ ` ! @ # \$ % ^ & * () _ - + = [] { } | \ ; : ' " , < . > ? /

Note that for a long time PreTeXt had empty elements for many of these characters, as a consequence of naïveté. So you might see `<dollar/>`, `<ampersand/>`, or others in old source. They will be deprecated and will raise warnings.

Now, when a character is nowhere to be found on your keyboard, we provide conveniences as markup. Or a keyboard character may have a different variant which we implement as an empty element. Here we test many of these. Read the Author's Guide for tags and more detail.

© ® ¢ ® ™ ™ ... · ~ ‰ ¶ § − × ÷ ±

There are a few common abbreviations of Latin phrases that can be achieved in HTML one way, and in L^AT_EX with a slightly different mechanism. These are due to L^AT_EX's treatment of a period (full stop), depending on its surroundings. So not reserved characters, but just divergent treatment. Using these will lead to the best quality in all your outputs. See Will Robertson's informative and arcane [blog post](#) on the topic if you want the full story for the treatment of a full stop in L^AT_EX.

Tag	Realization	Meaning
ad	AD	<i>anno Domini</i> , in the year of the Lord
am	AM	<i>ante meridiem</i> , before midday
bc	BC	English, before Christ
ca	ca.	<i>circa</i> , about
eg	e.g.	<i>exempli gratia</i> , for example
etal	et al.	<i>et alia</i> , and others
etc	etc.	<i>et caetera</i> , and the rest
ie	i.e.	<i>id est</i> , in other words
nb	NB	<i>nota bene</i> , note well
pm	PM	<i>post meridiem</i> , after midday
ps	PS	<i>post scriptum</i> , after what has been written
vs	vs.	<i>versus</i> , against
viz	viz.	<i>videlicet</i> , namely

We also distinguish between abbreviations (vs.), acronyms (SCUBA) and initialisms (XML). This is a test of the text version of a multiplication symbol: 2 × 4.

Simple coordinates with degrees, minutes, seconds, or temperature, or distance in feet and inches. “We parked the car at 36°16'0.83"N, 122°35'47.27"W, and since it was 93°F, we walked 505'3.6" so we could swim in the bay.”

An **em dash** is the long dash used much like parentheses (not an **en dash** used to denote a range, such as a range of page numbers). It should not have spaces around it, but some style guides allow for a *thin* space, which—we test right now. A publication file entry can be set to `none` or `thin` to control this.

9.5 Currency

For best results, be certain the right Unicode characters are in your source. If you only need a certain symbol rarely, you can enter it in your source via its Unicode number. For example, to obtain a peso, type `₱`. This table has been tested with our default fonts, and should be fine for HTML output. Please

report any difficulties with different $\text{\LaTeX}$ fonts, as there are extra measures we can take to make these more robust. (We’ve already done this for the Paraguayan guaraní.)

Table 9.2 Supported Currency

Sign	Unicode	Name
\$	U+0024	dollar
¢	U+00A2	cent
£	U+00A3	sterling
₡	U+00A4	currency
¥	U+00A5	yen
f	U+0192	florin
฿	U+0E3F	baht
₯	U+20A1	colon
₺	U+20A4	lira
₮	U+20A6	naira
₩	U+20A9	won
₫	U+20AB	dong
€	U+20AC	euro
₱	U+20B1	peso
	U+20B2	guarani

9.6 Icons in Text

A limited supply of icons can be used when explaining how to use some computer application. The empty element is `<icon/>` and the attribute is `@name`.

We sprinkle  a few into a few sentences  to check baselines  and font sizing. We sprinkle  a few into a few sentences  to check baselines  and font sizing. We sprinkle  a few into a few sentences  to check baselines  and font sizing. We sprinkle  a few into a few sentences  to check baselines  and font sizing.

Table 9.3 User-Interface Icons

Name	Icon	Name	Icon	Name	Icon
arrow-down	↓	arrow-left	←	arrow-right	→
arrow-up	↑	file-save		gear	
menu	≡	wrench			

Table 9.4 Creative Commons License Icons

Name	Icon	Name	Icon	Name	Icon
cc		cc-by		cc-sa	
cc-nc		cc-pd		cc-zero	

Nominations of new icons must

- Have a Unicode character representation.
- Be in the HTML/CSS/JS Font Awesome 5 catalog.
- Be in the $\text{\LaTeX}$ fontawesome5 package.
- Have a reasonably semantic PreTeXt name.

Please supply all this information, including the official Unicode name, with your request. Better yet, form a pull request.

Warning 9.5 Icons, xelatex, and Fonts. When processing a $\text{\LaTeX}$ file with `xelatex` the Font Awesome 5 icons are expected to be in a *system* font whose name is Font Awesome 5 Free. This is not a filename, and installing the $\text{\LaTeX}$ `fontawesome5` package into your $\text{\LaTeX}$ installation does not always guarantee that this font will automatically be available as a *system* font.

The Publisher’s Guide contains some discussion about installing fonts into a system, as part of the documentation of creating a $\text{\LaTeX}$ style, and has particular warnings about only using the $\text{\LaTeX}$ `fontawesome5` package as a vehicle for installing and accessing these fonts.

9.7 Keyboard Keys

Your text can include specialized text meant to look like a key on the keyboard of a calculator or other device. So you can go `b` `Enter` `<` or `F1`. Or maybe a sequence as: `Tab` `>` `Ctrl` `>` `T`. Use the `<kbd>` element, with the label of the key as content.

There is a growing supply of keys which are labeled with graphics rather than text, such as a left arrow `<`, right arrow `>`, up arrow `↑`, down arrow `↓`, and Enter `↵`. See *The PreTeXt Guide* for the definitive list. In `??` the literal column means the symbol/character is the content of a `<kbd>` element, while the named column means the symbol/character has been chosen via the value of the `@name` attribute of an empty `<kbd/>` element.

Table 9.6 Named keys

	Literal	Named
Ampersand	<code>&</code>	<code>&</code>
Less than	<code><</code>	<code><</code>
Greater than	<code>></code>	<code>></code>
Dollar	<code>\$</code>	<code>\$</code>
Per cent	<code>%</code>	<code>%</code>
Open brace	<code>{</code>	<code>{</code>
Close brace	<code>}</code>	<code>}</code>
Hash	<code>#</code>	<code>#</code>
Backslash	<code>\</code>	<code>\</code>
Tilde	<code>~</code>	<code>~</code>
Circumflex	<code>^</code>	<code>^</code>
Underscore	<code>_</code>	<code>_</code>

Table 9.7 Upper Case

<code>~</code>	<code>!</code>	<code>@</code>	<code>#</code>	<code>\$</code>	<code>%</code>	<code>^</code>	<code>&</code>	<code>*</code>	<code>(</code>	<code>)</code>	<code>_</code>	<code>+</code>	
<code>Tab</code>	<code>Q</code>	<code>W</code>	<code>E</code>	<code>R</code>	<code>T</code>	<code>Y</code>	<code>U</code>	<code>I</code>	<code>O</code>	<code>P</code>	<code>{</code>	<code>}</code>	<code> </code>
<code>CapsLock</code>	<code>A</code>	<code>S</code>	<code>D</code>	<code>F</code>	<code>G</code>	<code>H</code>	<code>J</code>	<code>K</code>	<code>L</code>	<code>:</code>	<code>'</code>	<code>Enter</code>	
<code>Shift</code>	<code>Z</code>	<code>X</code>	<code>C</code>	<code>V</code>	<code>B</code>	<code>N</code>	<code>M</code>	<code><</code>	<code>></code>	<code>?</code>	<code>Shift</code>		

Table 9.8 Lower Case

<code>`</code>	<code>1</code>	<code>2</code>	<code>3</code>	<code>4</code>	<code>5</code>	<code>6</code>	<code>7</code>	<code>8</code>	<code>9</code>	<code>0</code>	<code>-</code>	<code>=</code>	
<code>Tab</code>	<code>q</code>	<code>w</code>	<code>e</code>	<code>r</code>	<code>t</code>	<code>y</code>	<code>u</code>	<code>i</code>	<code>o</code>	<code>p</code>	<code>[</code>	<code>]</code>	<code>\</code>
<code>CapsLock</code>	<code>a</code>	<code>s</code>	<code>d</code>	<code>f</code>	<code>g</code>	<code>h</code>	<code>j</code>	<code>k</code>	<code>l</code>	<code>;</code>	<code>'</code>	<code>Enter</code>	
<code>Shift</code>	<code>z</code>	<code>x</code>	<code>c</code>	<code>v</code>	<code>b</code>	<code>n</code>	<code>m</code>	<code>,</code>	<code>.</code>	<code>/</code>	<code>Shift</code>		

9.8 URLs, such as `http://example.com`

The `<url>` element can be used to create an external reference. The mandatory `@href` attribute is the actual URL complete with the protocol (e.g. `https://`). Content for the element is optional, and if provided will be the “clickable” text. In this case, a `@visual` attribute can be provided. In producing the print version of a PDF, this will become a parenthetical with a more friendly version of the URL. When no content is provided, the “clickable” will be the URL with a preference for an optional `@visual`. This subsection has some (extreme) tests and we leave complete documentation and full details for the PreTeXt Guide.

A long URL for testing: www.pcc.edu/enroll/registration/dropping.html#withdraw. Notice in the source that you *do not* put any tags inside the @href or @visual attributes, but you may need to provide XML escape sequences (see Subsection ??).

A `<url>` element in the print version of a PDF will get a trailing parenthetical containing the (simplified) URL in the highly-recommended `@visual` attribute. If you do not provide the `@visual` attribute in this case, then you will get the `@href` value repeated, possibly with some editing. If you insist, you can make the `@visual` attribute identical to the `@href` attribute. Some tests:

- With a useful @visual attribute
- With a duplicate @visual attribute
- With no @visual attribute, so an edited one is formulated (no protocol)

Here is a totally bogus URL, which contains every possible legal character, so if this fails to convert there is some problematic character. In order to test the use of a percent sign (%) in a URL, we follow it by two hex digits, specifically, 58, which is a way to represent the character X in a URL. Normal text, monospace text, <url> with just @href, <url> with @href and @visual, <url> with @href, @visual and content. Notice how the various versions do, or do not, line-break in L^AT_EX/PDF output, including the (potentially confusing) use of a use of a hyphen in the normal text version. Note that # starts a fragment identifier, so it is the penultimate character. A ? would normally start a query string, but as the fragment identifier, it is not special. Finally, square brackets ([and]) are only legal in limited situations (authority component for IPv6 host literals), so they are present as percent-encoded literals, which really does not demonstrate anything. All these efforts to make a comprehensive, but legal, URL are meant to appease the EPUB validator which will otherwise flag them as errors.

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789%-
 . _ ~ : / % 5 B % 5 D @ ! \$ & ' () * + , ; = # ?
 ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789%-
 . _ ~ : / % 5 B % 5 D @ ! \$ & ' () * + , ; = # ?
[http://example.com/ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789%58-. _ ~ : / % 5 B % 5 D @ ! \\$ & ' \(\) * + , ; = # ?](http://example.com/ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789%58-. _ ~ : / % 5 B % 5 D @ ! $ & ' () * + , ; = # ?)
[example.com/ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789%58-. _ ~ : / % 5 B % 5 D @ ! \\$ & ' \(\) * + , ; = # ?](http://example.com/ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789%58-. _ ~ : / % 5 B % 5 D @ ! $ & ' () * + , ; = # ?)
 Characters to a footnote

Line-breaking in L^AT_EX/PDF output is specialized, using path separators (slashes) as candidates for splitting across lines:

[http://example.com/long/long/long/long/
long/long/long/long/long/long/long/long/long/long/long/long/long/longer/
longer/longer/longer/longer/longer/longer/longer/longer/longer/longer/longer/](http://example.com/long/long/long/long/long/long/long/long/long/long/long/long/long/longer/longer/longer/longer/longer/longer/longer/longer/longer/longer/)

We are not fans of footnotes, they are totally unstructured.¹ A URL in a footnote migrates around, and so care must be taken.² This paragraph has two footnotes, one with a real URL from Jesse Oldroyd, another with a fake URL from the above suite. For good measure, we repeat the URL found in the first footnote³: [Carleson's Theorem](#). And we include a no-content version of the same link, with a visual version provided and employed: [en.wikipedia.org/wiki/Carleson%27s_theorem](#).

There is an attribute, @ncbi that you can use on the `taxon` element to precisely identify the organism you are discussing using an identification number from the [National Center for Biotechnology Information](http://www.ncbi.nlm.nih.gov/taxonomy). Their `taxonomy` is at www.ncbi.nlm.nih.gov/taxonomy. Right now, we do not do anything with this attribute, but things like links are certainly possible. See the source of this document to see it in use with *Drosophila miranda* which could be used to construct a link to [further information via id number](#) or even [further information via just the name](#).

(u, Univariate Polynomial Ring in u over Integer Ring)

Ampersands, less-than, and greater symbols are likely to be necessary in source code, such as Sage code (think generators of field extensions) or TikZ code (think arrowheads), and in matrices (think separating entries). If you have a big matrix, or a huge chunk of TikZ code, you can protect it all at once from the XML processor by wrapping it in `<![CDATA[ ]]>`. It should be possible to write without ever using the “CDATA” mechanism, but it might get tedious in places to use the supplied macros or XML escape sequences. This construction is often mis-understood as a solution better remedied by reading Subsection ?? again.

We test the three pre-defined L^AT_EX macros for `&`, `<`, and `>` with a pair of aligned equations:

$$\begin{aligned} a^2 + b^2 &< c^2 \\ c^2 &> a^2 + b^2 \end{aligned}$$

9.12 Jupyter Notebook, Markdown, MathJax, Delimiters

A Jupyter notebook allows a mix of HTML (our logistical preference for a conversion) and Markown (another set of special characters and their escaped versions). Certain pairs of delimiters, when appearing in consecutive HTML `<code>` elements require extraordinary care. But the one nut we cannot crack is pairs of dollar signs. So the next paragraph is known to render badly in a Jupyter notebook, but should otherwise be a bit boring.

\$ and \$

9.13 L^AT_EX Characters, Ligatures, and More

This section is just for testing, and the more you know about L^AT_EX, the more we would encourage you to *not* to read this. Look to the Author’s Guide for the *right* way to author your source.

The ten reserved characters, directly in the source: `# $ % ^ & _ { } ~ \`. And again: `X#X$X%X^X&X_X{X}X~X\X`, but smashed up tight to intermediate characters.

In a verbatim presentation: `# $ % ^ & _ { } ~ \`.

And `X#X$X%X^X&X_X{X}X~X\X`. (These verbatim versions are authored in different paragraphs to work around the Jupyter notebook bug described above.)

We also disrupt certain constructions from L^AT_EX. Attempting to sneak-in any traditional macro for the purposes of L^AT_EX-only output, such as, say a `\newpage`, will fail since the leading backslash will be caught and converted to `\textbackslash`. (See? It just happened twice.) For technical reasons we want to particularly test `\textbackslash`, `\textbraceleft`, and `\textbraceright`.

Four “T_EX ligatures”, `--`, `---`, ````, and `'`, authored in running text, `--`, `---`, ````, `'`. It may be hard to tell that the two consecutive apostrophes have not coalesced into a curly left smart quote, but see below, the spacing is subtly different.

We want the double quote mark from your keyboard, `"`, to not morph into some other character: `"`.

More testing: runs of hyphens. Such as: `-` (one), `--` (two), `---` (three), `----` (four), `-----` (five), `-----` (six), `-----` (seven). Use the empty elements `<ndash/>` and `<mdash/>` for the longer dashes/hyphens.

Runs of apostrophes should not become smart right double quotes: ' (one), '' (two), ''' (three), '''' (four), ''''' (five), '''''' (six), ''''''' (seven). You might want to cut-and-paste these into a text file to convince yourself there are the right number of characters. Here are two smart right double quotes, separated by a non-breaking space, for visual comparison: " ". Or 30 apostrophes on a line of their own (longer) followed by 15 smart right double quotes (shorter).

Runs of backticks (accent grave) should not become smart left double quotes when the output is processed by L^AT_EX: ``` (one), ```` (two), ````` (three), `````` (four), ```````` (five), `````````` (six), ```````````` (seven). Furthermore, in a context where Markdown syntax is recognized as well (e.g. a Jupyter notebook), paired backticks should *not* produce ``inline verbatim text``.

The next paragraph has a long run of words separated/joined by the keyboard forward-slash character. With this input, L^AT_EX will not line-break at the slash, nor will it hyphenate anywhere. PreT_EXt automatically provides an improved slash, which will line-break, as you should see below in L^AT_EX output. There is a bad right margin, but that is due to the absurdity of this test. This sort of problem should be no better or worse for the use of this character. Further refinements (zero-width space) and packages can be used to get hyphenation. HTML will line-break rationally with no extra help. Remember the `<solidus>` character for super-simple text fractions like 7/32 (which will *not* line-break), and math elements or SI unit markup for technical work.

A/test/of/some/short/words/that/go/off/the/end/of/a/line/A/test/of/some/short/words/that/go/off/the/end/of/a/line/A/test/of/some/short/words/that/go/off/the/end/of/a/line/A/test/of/some/short/words/that/go/off/the/end/of/a/line/A/test/of/some/short/words/that/go/off/the/end/of/a/line.

9.14 HTML and accidental mathematics

We render mathematics in web pages with the fantastic MathJax Javascript library. Simplifying just a bit, it recognizes L^AT_EX syntax within a page, takes control of that text, and replaces it with nice fonts and formatting. Now, if you write about L^AT_EX you might well have some mathematics in your examples. Best practice would be to use verbatim text for that, and we mark off such text as being off-limits to MathJax.

But if you are writing running text, then you can (accidentally) author some text which MathJax recognizes and converts to something (unintended). And if you are doing this intentionally, then you have ignored PreTeXt markup for mathematics, and are missing out on some features.

A few tests that we can prevent any accidents.

Inline mathematics: (x^2) .

Display mathematics:
$$x^2+y^2=z^2$$

Double backticks is a common L^AT_EX construction, which in L^AT_EX/PDF output *should not* become an opening quote-mark. Also, a single backtick in HTML is a signal for MathJax to interpret ASCII_{Math}, and then a double backtick causes “random” pieces of mathematics on a page to not render at all. So we have a quotation authored L^AT_EX-style: ``We have nothing to fear, but fear itself.”

10 Graphics

In addition to including images created externally (e.g. photographs), PreTeXt supports several languages for describing diagrams and pictures with human-readable source code (i.e. plain text), rather than using a “paint” program. This section describes the various methods for incorporating, or generating, graphics, images or diagrams.

10.1 Images from External Sources

If you have raster images (photographs, etc.) then they are specified with complete filenames, as above in Figure ?? or just below.



Figure 10.1 New Zealand Landscape, commons.wikimedia.org, CC-BY-SA-2.0

If you have existing images that are vector graphics, then PDF format works best for $\text{\LaTeX}$ output and SVG format works best for HTML. The utility `pdf2svg` works well for converting PDF to SVG. In this case, specify your source as a filename, but leave off the file extension, and the appropriate version will be used for the current output format.

The image below is provided from a PDF file in $\text{\LaTeX}$ output, and was converted to an SVG for use with the HTML output. It has been explicitly scaled to a width of 65% of the text width. It has a `<description>`, but no `<shortdescription>`, so is testing that scenario.

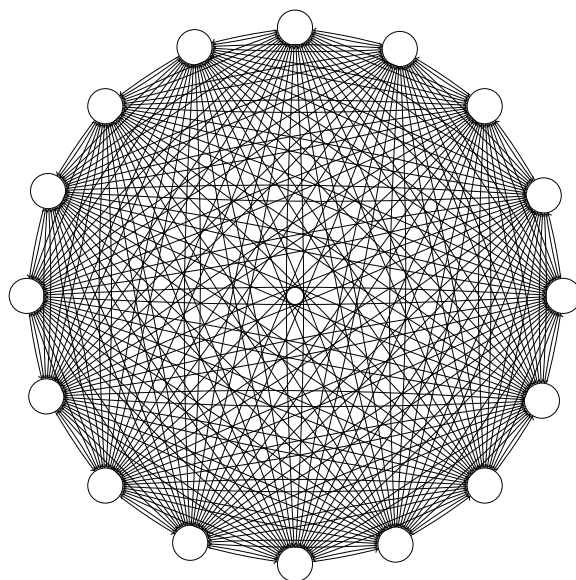


Figure 10.2 Complete graph on 16 vertices, from www.texample.net

Remark 10.3 Footnote Buried. Nested `tcolorbox` (in $\text{\LaTeX}$ conversion) need special care when footnotes are interior.

A paragraph A second para-
interior to a graph, just to avoid
sidebyside with a one-panel warn-
footnote¹ buried in-
side the paragraph.

The final paragraph of this remark, randomly placed, to test footnotes in $\text{\LaTeX}$ conversions.

10.2 PreFigure

PreFigure is a standalone project for authoring mathematical diagrams (see prefigure.org). Its philosophy and approach are much like that of PreTeXt, and PreFigure is tightly integrated into PreTeXt. One key feature is excellent support for the creation of accessible output formats.

As of 2024-11-06 development continues for PreFigure itself, and fine-tuning of its integration within PreTeXt. But it is usable now for projects that want to use it.

- You can author PreFigure diagrams, and then generate SVG, PDF, and PNG output versions with the `pretext/pretext` script (see the PreTeXt Guide), and expect them to render in HTML, PDF, and EPUB output formats.
- Look for automatic generation to come to the PreTeXt-CLI sometime very soon.
- PreFigure has excellent support for annotations within an SVG diagram, supporting their use by screenreaders, for example. See an example below.

¹Interior footnote.

- Production of tactile versions is now possible, though they are not incorporated explicitly into any of the output formats. Perhaps they will become available via archive links or as a zip archive.

This is a basic PreFigure diagram, in the sense that it does not use all the features of PreFigure or PreTeXt.

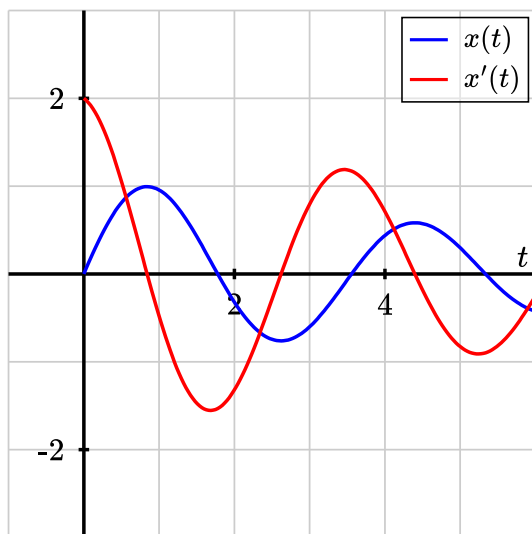
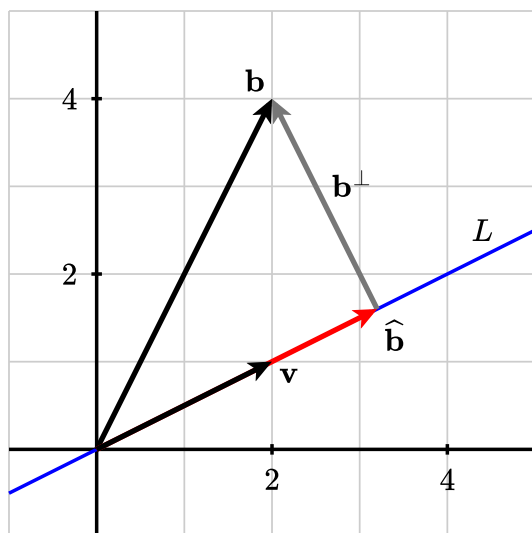


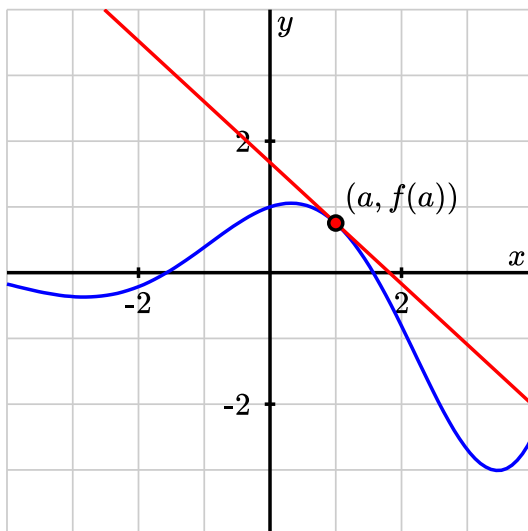
Figure 10.4 Solution to a differential equation

This next diagram employs some $\text{\LaTeX}$ macros that are defined in the usual way in `<docinfo>` and are employed to produce the names of some vectors in the labels. The blue line is colored blue by a global PreFigure declaration, also in `<docinfo>`.



The next PreFigure diagram is authored with annotations, arranged in a hierarchy of increasing refinement and detail. Each identified graphical component will read its annotation and show it on the screen below the diagram. When a reader clicks on the image, a high-level summary will be read using the author-provided annotation. The down and up arrow keys enable a reader to explore the diagram in more or less detail while the right and left arrow keys reveal features at the same level of detail. When the focus is on the graph,

pressing "O" will produce a sonification of the graph.



Including annotations enables a new type of interactive diagram within a PreTeXt document offering potential benefits for all readers. In particular, annotations allow an author to call the reader's attention to specific details in a diagram and how they are related to one another so that the diagram and surrounding text are more tightly integrated. The annotated diagram below introduces Fibonacci tilings, which are one-dimensional analogs of Penrose tilings, and offers an explanation of their aperiodicity. Of course, surrounding text would usually provide a richer context for a diagram like this.

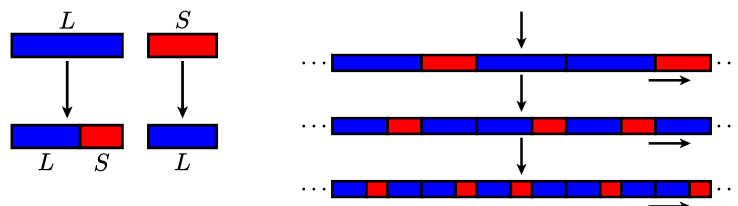
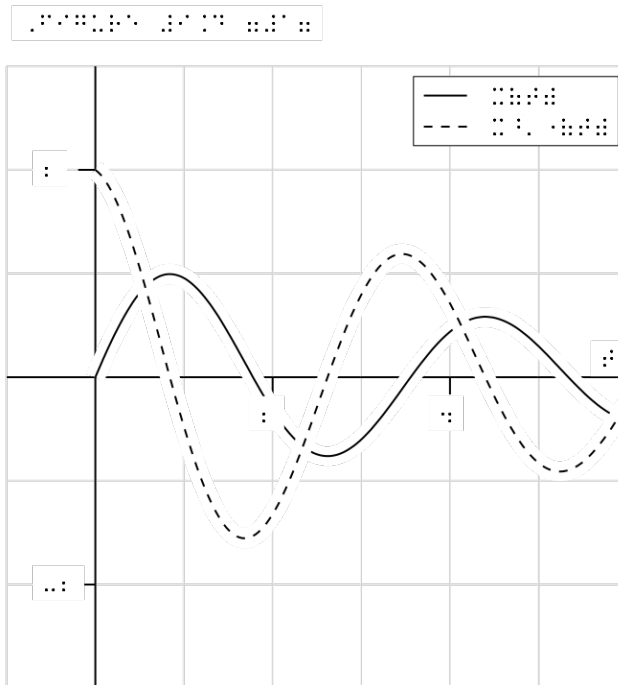


Figure 10.5 Fibonacci tilings are one-dimensional analogs of Penrose tilings. The diagram on the left introduces the process of deflation that is used to produce tilings while the diagram on the right explains why they are aperiodic.

For sighted readers, here is an example of a tactile version of one of the above diagrams. It is generated automatically from the same source as the other version. Imagine this being “printed” with an embosser so that the parts of the diagram, and the braille labels, are raised up from the paper and can be explored with one's fingertips. This image is a PNG produced specifically for this document. Typically, a tactile diagram produced by PreFigure will be a PDF ready to be sent to an appropriate embosser. Notice that PreTeXt adds a caption indicating the diagram's location in the document along the top of the diagram.



10.3 $\text{\LaTeX}$ images

There are several graphics engine packages that a $\text{\LaTeX}$ document can employ. Code from these packages renders diagrams automatically as part of normal processing of $\text{\LaTeX}$ files. For HTML output the `pretext` script produces SVG versions of the pictures. The script can also produce standalone $\text{\TeX}$ source files, PDFs, PNGs, and EPSs. The packages should be loaded in `docinfo/latex-image-preamble`, which is also where global package settings should be made. If any ampersands occur in your $\text{\LaTeX}$ code you should use the `\amp` macro pre-defined by PreTeXt. These first examples are from the [TeXample.net](http://www.texample.net) site. Note that any $\text{\LaTeX}$ macros used in the rest of your document may be employed in the $\text{\LaTeX}$ -standalone or Asymptote diagrams (with this feature coming to Sage graphics next?).

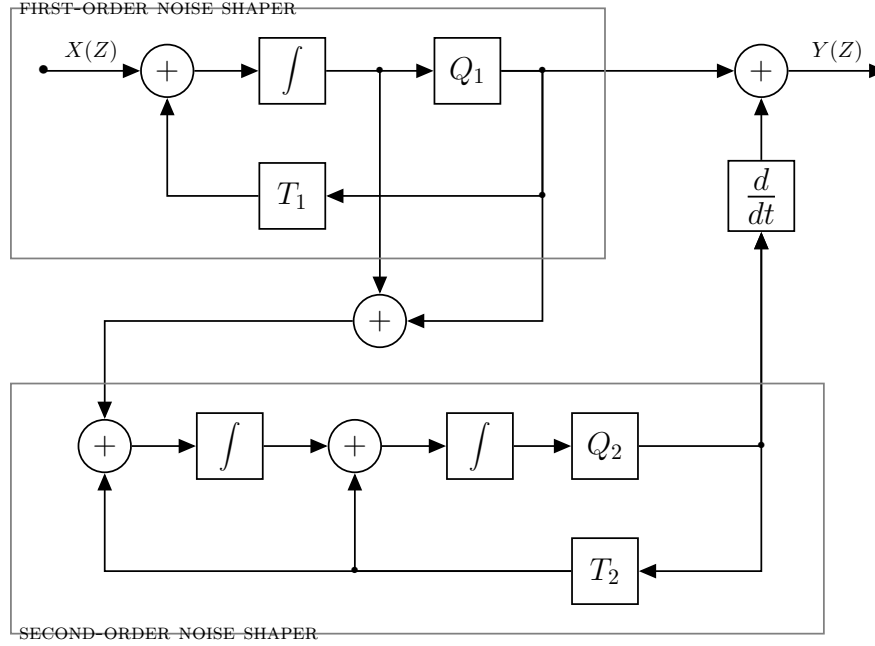


Figure 10.6 TikZ Electronics Diagram

The next example began life in [Sketch](#), which will output TikZ code (though the code has been edited by hand for readability).

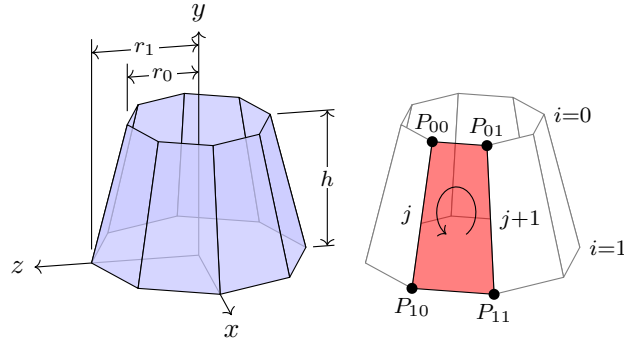


Figure 10.7 TikZ Cone Drawing

The `pgfplots` package was included in `docinfo/latex-image-preamble`. Here, it is used. Also, here we demonstrate using `\amp` where you would normally use an ampersand in $\text{\LaTeX}$. There are known issues with `xelatex` processing any gradient shading in `tikz`. To (successfully) create the gradient shading in the 3D image here, you may need to use `pdflatex` until $\text{\LaTeX}$ developers resolve this issue.

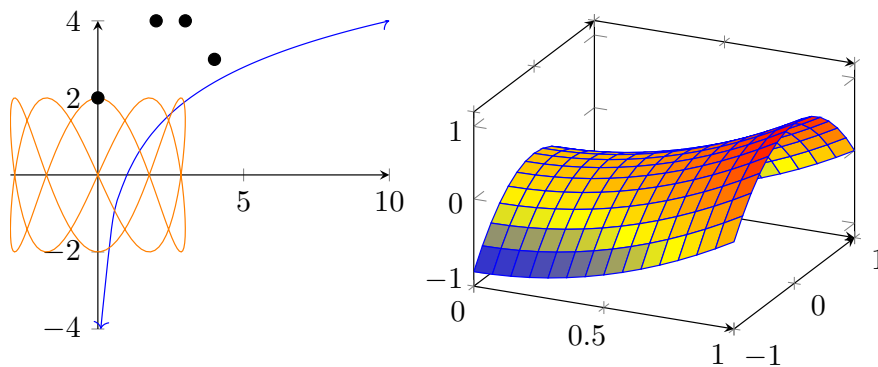


Figure 10.8 Sample pgfplots plot

A plot might use a graphics language to draw the axes and grid, but the data might be from an experiment and live in an external file that you do not wish to place in your source. Place such a file in a subdirectory directly below the directory where your master source file resides. Then indicate this directory in a `docinfo/directories/@data` attribute of your source. But you *must* prefix the path with `data/` as in the source below.

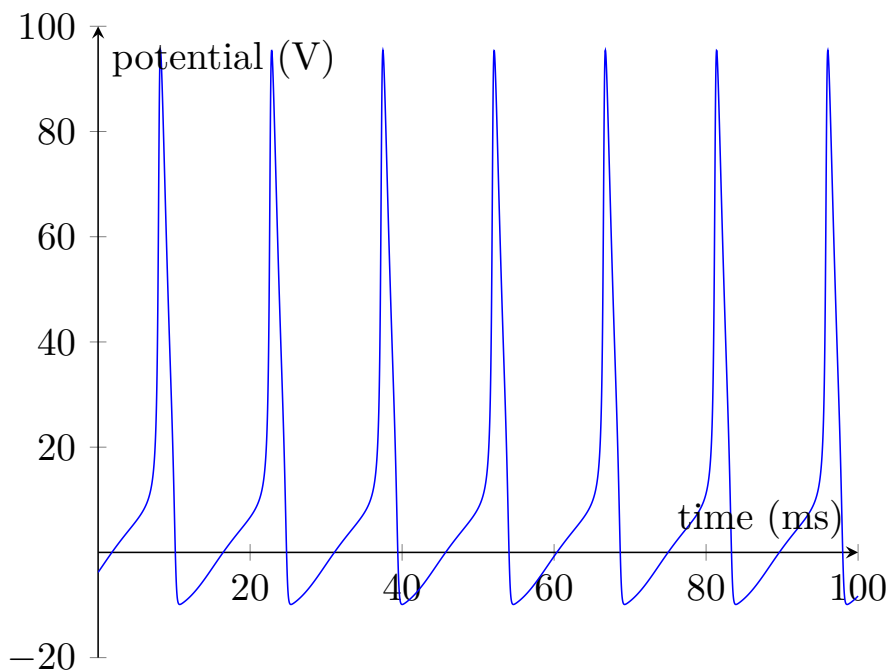


Figure 10.9 External data in a pgfplots plot

The TikZ image in the next figure is made up from two PNG images (the shark and the swimmer), in addition to various TikZ commands. The images reside in a source directory, `numerical/attack`, so the image files (`shark.png`, `swimmer.png`) are prefixed in the TikZ code with `data/attack` so the creation of the image will be successful. This example is courtesy of Stephen Brown.

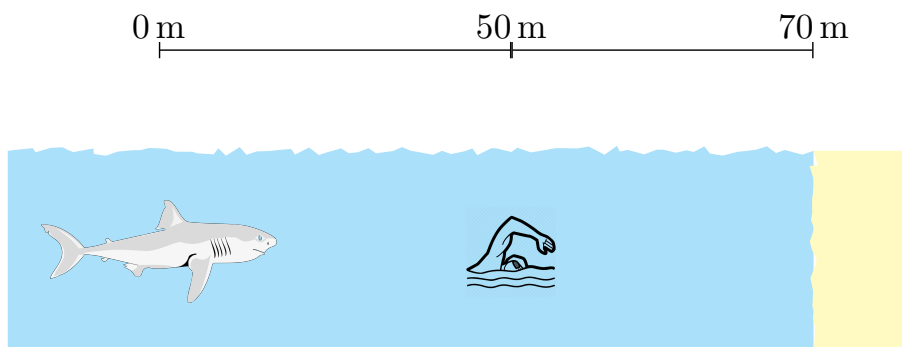


Figure 10.10 An illustration of the shark trying to catch you as you swim to shore.

A Cartesian plot might benefit from having a `<description>` with a `<tabular>` that lays bare the data used to to plot points.

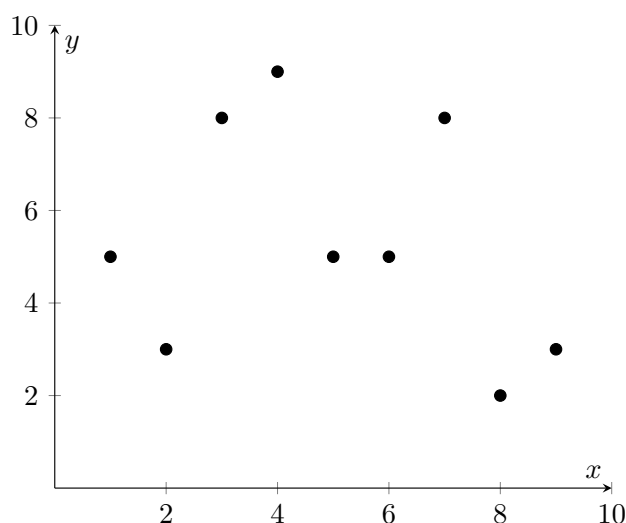


Figure 10.11 Full description with tabular

The next image requires three passes with $\text{\LaTeX}$ to get everything in place. It is placed here to test that the code in the Python script correctly recognizes this requirement.

$$\begin{bmatrix} 3 & 4 & 0 & 1 & 4 \\ -2 & 3 & 1 & -2 & 3 \\ 0 & 2 & 1 & 0 & 2 \end{bmatrix}.$$

Figure 10.12 A matrix with colored entries

PSTricks is a $\text{\LaTeX}$ package for drawing diagrams and pictures, dating back to the days before PDF, when PostScript (PS) was king. Given its history, it does not seem to work easily with the `pdf $\text{\LaTeX}$` engine. But it will work easily with the `x $\text{\LaTeX}$` engine. We try to keep this present sample document workable with both engines, so we have presented an example of the use of PSTricks in the `x $\text{\LaTeX}$` -exclusive sample document where we test obscure fonts and characters. So your best bet is to look there.

There are suggestions online that

```
\usepackage[pdf]{pstricks}
```

along with

```
pdflatex --shell-escape *.tex
```

is workable. We could not make it happen, and a “shell escape” can be a dangerous security hole. That said, updates to this approach are welcome.

10.4 Asymptote, 2D

The Asymptote graphics language may be placed in your source to draw graphs, diagrams or pictures. Rules for formatting code are identical to those for Sage code. For more on Asymptote see asymptote.sourceforge.net.

This is a simple physics diagram about levers, taken from the Asymptote documentation. In the HTML version of this article, the images are SVG’s and so should scale nicely when you zoom in on the page.

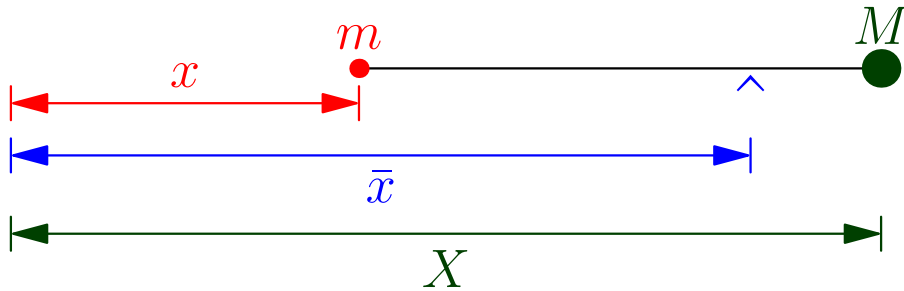


Figure 10.13 Asymptote Lever Demonstration

And a colorful contour plot with logarithmic scale. Again, from the Asymptote documentation. This SVG image employs two additional PNG images for the two parts where the color varies continuously.

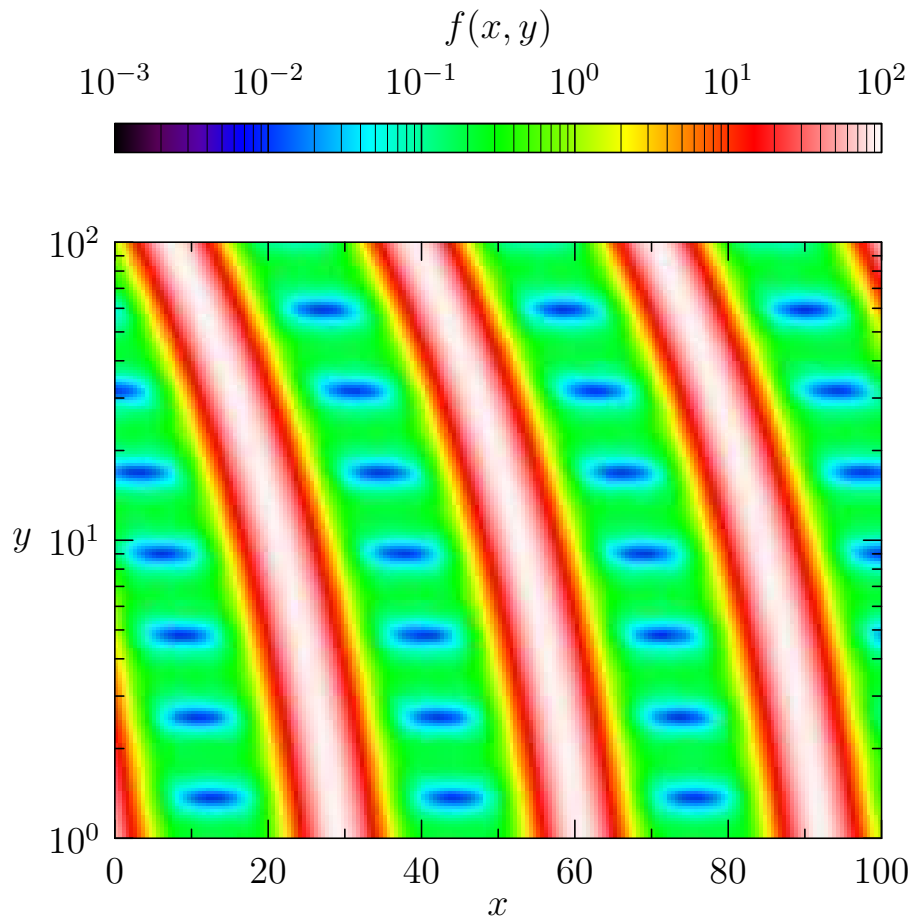


Figure 10.14 Asymptote Contour Plot

Here is the lever diagram again, but now we have added an integral to one of the legends, *using a $\text{L}^{\text{A}}\text{T}_\text{E}^{\text{X}}$ macro of our own*, which is identical to one we used in the early part of this article. The point is, we only needed to define the macro once for the entire document, and it is available as we make Asymptote diagrams. This device can be used to maintain flexibility and consistency in your choice of notation.

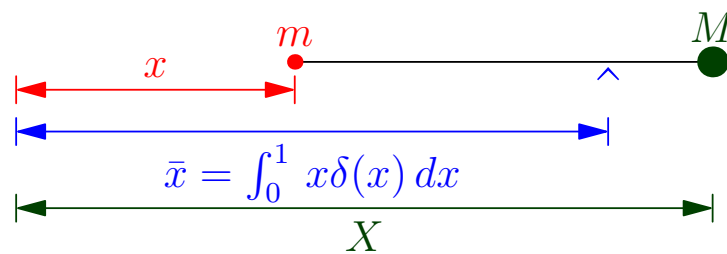


Figure 10.15 Asymptote Lever, plus Integral

10.5 Asymptote, 3D via WebGL

Asymptote can create an HTML file that is an interactive version of a 3D shape. At this writing (2020-05-18) support via the `pretext` script is evolving. Plus, you will need newer versions of Asymptote and the `dvisvgm` utility to duplicate all of the results being displayed here in this testing document. The other distinction is that the author needs to provide the aspect ratio of the figure,

and this should be placed on the `<asymptote>` element (not on the `<image>` element). Figure ?? is from the [Asymptote Gallery](#).

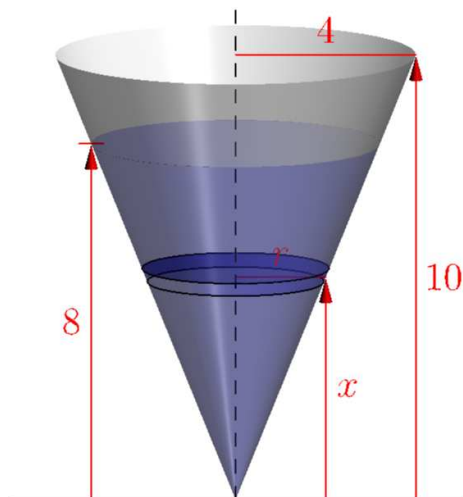


Figure 10.16 Work Cone (Asymptote Interactive 3D Image)

These 3D images in HTML output are rotatable with a pointing device (mouse, trackpad) with a click-and-drag. A finger should suffice on touch-sensitive devices (phones, tablets). Zooming in and out can be accomplished with a mouse wheel, or by pinching. As a contribution to the accessibility of PreTeXt HTML output, keyboard controls will also allow for exploration of these images. (Make sure the image has focus when you attempt to use these.)

Table 10.17 3D Image Keyboard Controls

Key	Action
x	Rotate around x -axis
y	Rotate around y -axis
z	Rotate around z -axis
+	Enlarge image
-	Shrink image
h	Return to home position

And finally, an example of a 3-D graph (from the Asymptote documentation again). This WebGL image is a beautiful example of a Riemann surface. As you rotate the image, notice how the reflection of the light source varies, along with the brightness of various regions of the surface. This example is accomplished with just 10 lines of Asymptote code.

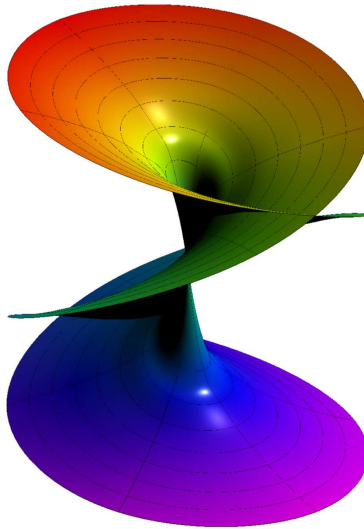


Figure 10.18 Asymptote 3-D Surface

10.6 Mermaid Diagrams

[Mermaid](#) is a Markdown-inspired tool for authoring various kinds of diagrams. Below, three of the available diagram types are demonstrated. For a full listing of diagram types, see the [Mermaid Documentation](#). The [Mermaid live editor](#) is a great tool for testing the syntax of your mermaid diagrams.

In PreTeXt, you can specify a Mermaid theme via the `@theme` in the `common/mermaid/` publisher variable.

For HTML output, if you switch back-and-forth between light-mode and dark-mode, you will need to refresh the page to see the changes in the Mermaid diagrams.

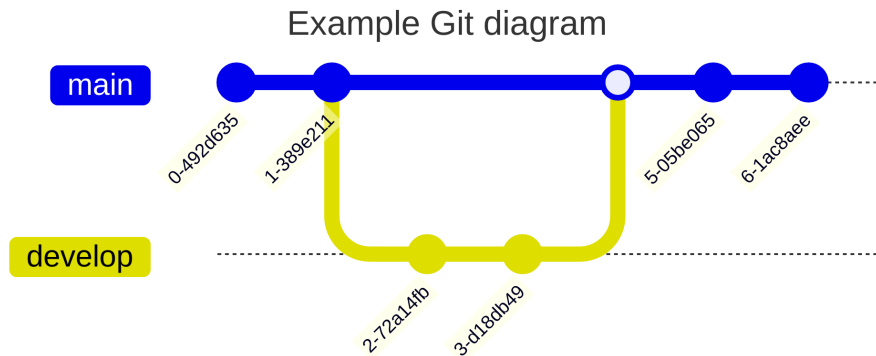


Figure 10.19 Mermaid Git Diagram

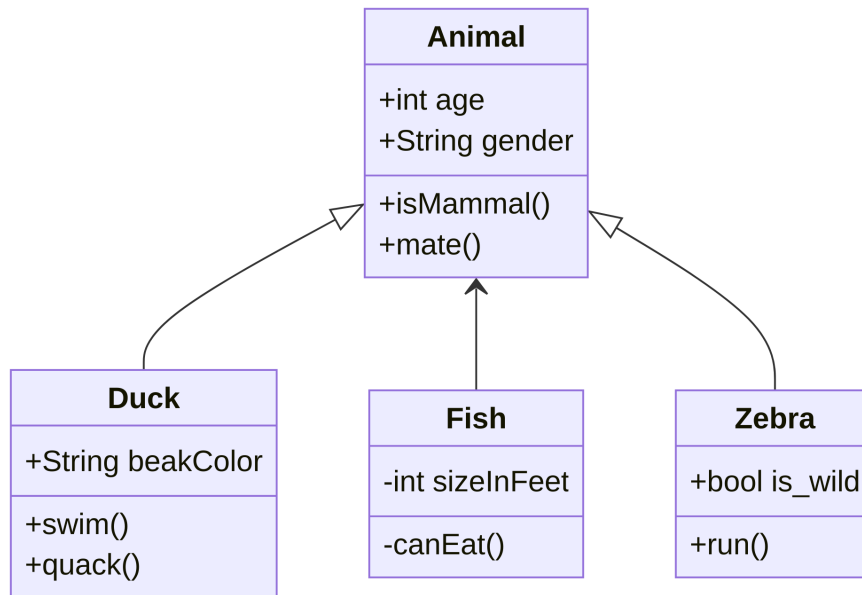


Figure 10.20 Mermaid Class Diagram

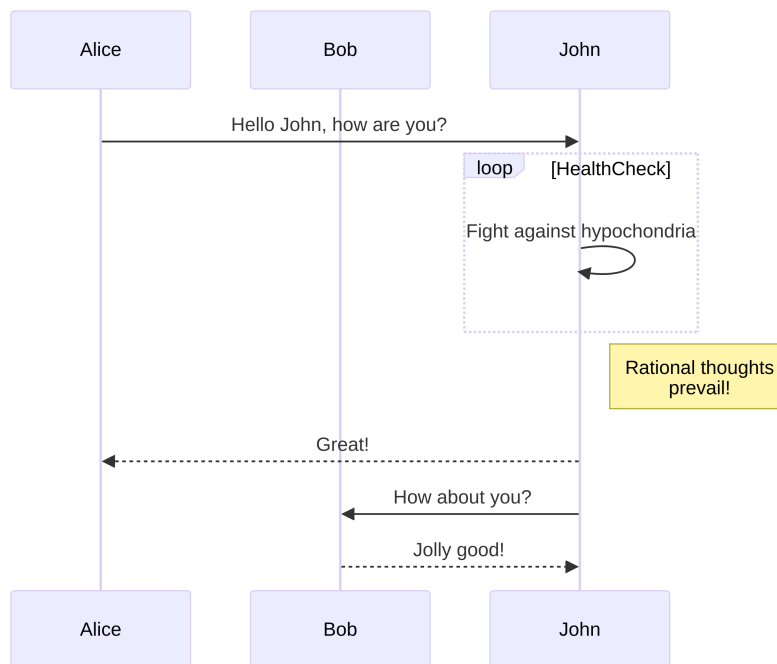


Figure 10.21 Mermaid Sequence Diagram

Mermaid has two layout engines—the default and `elk`. The ELK engine often does a better job with complex diagrams. You can specify it as a default by setting the publisher variable `<common/mermaid/@layout-engine>` to `"elk"`, or specify it in a single diagram by using standard Mermaid config frontmatter as shown in ?? below.

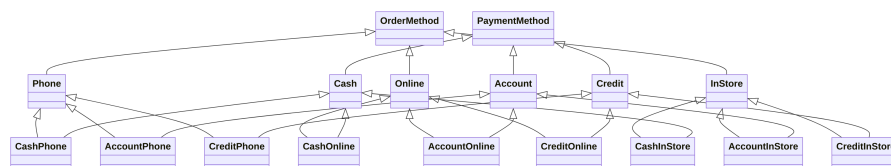


Figure 10.22 Mermaid Class Diagram using ELK layout engine

10.7 Sage Plots

Any of the numerous capabilities of Sage may be used to produce any graphics object, be it the simple graph of a single-variable function or some realization of a more complicated object. All of the usual rules about formatting Sage code (esp. indentation) apply, along with one more caveat. The last line of your Sage code *must* return a Sage Graphics object (or 3D plot). The `pretext` script will isolate this last line, use it as the RHS of an assignment statement, and the Sage `.save()` method will be called to generate the image, which is either a Portable Document Format (PDF) file amenable to $\text{\LaTeX}$ output, or a Scalable Vector Graphics (SVG) file amenable to HTML output. It is also possible to make a PNG image, which is necessary for an EPUB destined for a Kindle book. For visualizations of 3D plots, Sage will only produce Portable Network Graphics (PNG) files, which can be included in HTML pages or $\text{\LaTeX}$ output. For complete documentation, see the PreTeXt Guide as this subsection is not comprehensive.

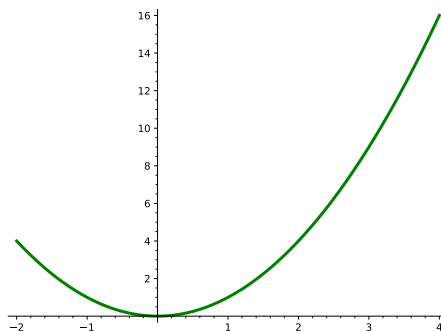


Figure 10.23 A Sage standard parabola, on $[-2, 4]$

Pay careful attention to the requirement that the last line of your code be a graphics object. In particular, while `show()` might appear to do the right thing, it evaluates to Python's `None` object and that is just what you will get. The code for Figure ?? illustrates creating two graphics objects and combining them into an expression on the last line that evaluates to a graphics object.

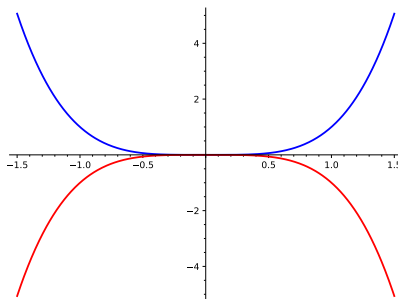


Figure 10.24 Two Sage plots on one set of axes

Sage code comprised of just a single line was once mishandled, leading to *no output*. From Jean-Sébastien Turcotte we have the example that revealed the problem.

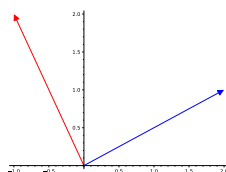


Figure 10.25 Les vecteurs $\vec{u}$ et $\vec{v}$

The following examples are from the [Sage Tour](#). We package them into a `sidebyside` layout element, see Section ??.

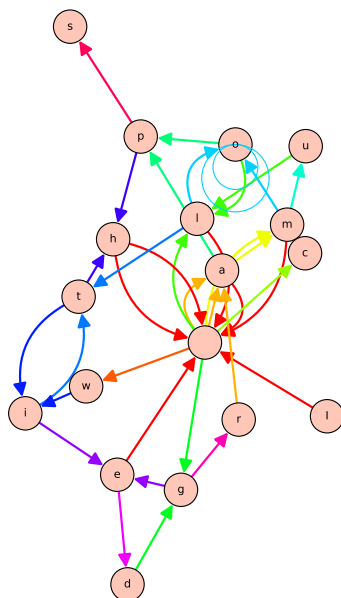


Figure 10.26 A Sage multi-graph of a sentence

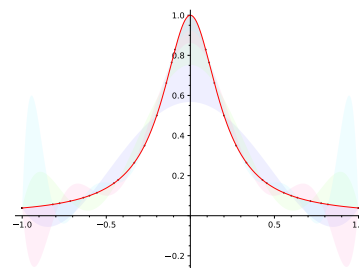


Figure 10.27 Sage polynomial approximations of $f(x) = 1/(1 + 25x^2)$

From the Sage documentation, with slight modifications, credited to Douglas Summers-Stay. A plot of the implicitly defined surface

$$2 = \cos(x + ty) + \cos(x - ty) + \cos(y + tz) + \cos(y - tz) + \cos(z - tx) + \cos(z + tx)$$

in rectangular xyz coordinates, with t equal to the golden ratio. If you set `plot_points=100` in the Sage code, you will get a very smooth rendering, but also a quite large HTML file. We have used `plot_points=50` to reduce the file size by a factor of four. Note the need for a value of `3d` for the `@variant` attribute, and an explicit aspect ratio with `@aspect`. Arrow keys, a mouse scroll wheel, plus grabbing with a left or a right mouse button, can be used to manipulate the image.

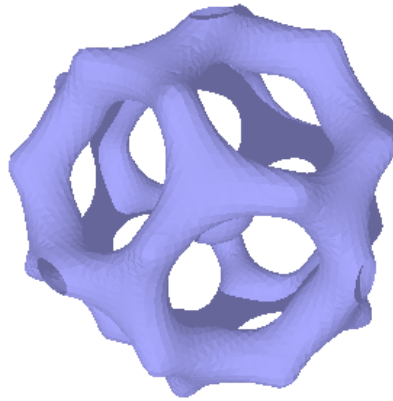


Figure 10.28 A Sage implicitly defined 3D surface

10.8 Inkscape Images

[Inkscape](#) is a great tool for creating images. It ticks all the boxes: open source, mature, cross-platform, standards-compliant. Read much more about it in [The PreTeXt Guide](#). In HTML output the two images below are both in SVG format. The first is “pure” SVG, while the second has embedded information that makes it easier to edit in Inkscape. You could view the source for this page in the HTML version, deduce the filename of the second image, download it, and manipulate it profitably with Inkscape. Both files are quite small, but the first is half the size of the second. In PDF the two images come from files that are identical, so nothing is being tested. The PDF version is smaller still.

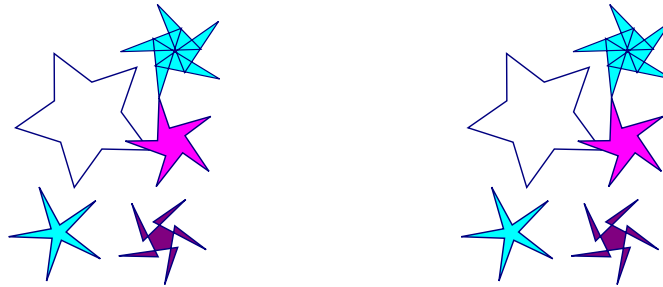


Figure 10.29 Inkscape Stars, Plain SVG (left), Inkscape SVG (right), from Bethany Llewellyn

10.9 Copies of Images

Sometimes you want to use the same image more than once. Here we just point to a PNG file that we repeat often throughout this sample.

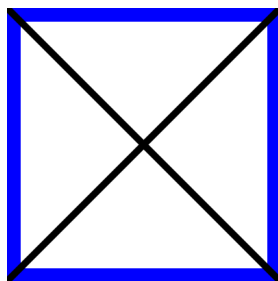


Figure 10.30 Copy of raster image, in a figure, so now numbered and captioned

For images described by code, such as TikZ code in a `<latex-image>` element, this is a bit subtler. See the PreTeXt Guide for a complete description. We also demonstrate this with the sample book, since it is all set up with the `xinclude` mechanism. See the two plots of the 8-th roots of unity in the complex numbers section of the chapter on cyclic groups.

10.10 Caption Testing

A caption could be as substantial as a paragraph, here we test out one such example.

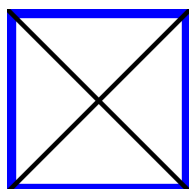


Figure 10.31 A caption can be a whole paragraph with lots of technical details, and maybe a hyperlink to something external, such as pretextbook.org or PreTeXt. There could be some inline mathematics, such as $x^2 + y^2 = c^2$. Would a knowl open here? Recursively? Let's see: ?? . Display mathematics, side-by-sides, theorems, and lots of other things should be banned. Footnotes sound like a bad idea. Strange characters should be fine: §.

10.11 Captionless Images

We strongly suggest placing images within a `<figure>`, as we have done above, so that you can reference them, and use the (required) `<caption>` to explain what they are. However there are places, such as a `<preface>`, where numbered items are not permitted. So you might want a solo image there. Or maybe graphics are an illustration of sorts, and a numbered figure feels like overkill. Or it is part of an `<exercise>` or `<proof>` of a `<theorem>`. But notice that you cannot then use this image as the target of a cross-reference, so you may need to refer to some enclosing container.

The image can be scaled by specifying the `@width` as a percentage, including the percent-sign (%). The height is scaled to preserve the aspect ratio. There is no facility to change the height, it is your responsibility to manage the aspect ratio independently. The `@margins` can be given as a pair of percentages, separated by a space. The `@width` defaults to 100%, while `@margins` defaults to the value `auto`, which will center the image. Missing values are computed sensibly, and there is robust error-checking. The layout control here is a subset of what is available for the more elaborate `<sidebyside>` element, see Section ??.

Two simple examples. The first has width 10% and so defaults to being centered, and the second has width 10% and left margin of 25%.



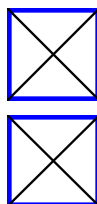
A paragraph, just to show where the first stops and the second ends.



You might wish to place a single image flush-left, or flush-right. You can specify the `margins` attribute as a pair of percentages for different left and right margins. The following are laid out with two margins, with a 0% left margin and right margin (respectively).

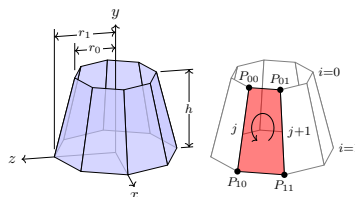


We place two images right above one another, to test spacing of consecutive images (provided they stay on the same page!).

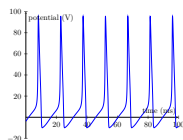


Testing (2019-06-02). All the images above are specified by filenames. We need to test how various options behave when incorporated into the (new) implementation for images, being introduced with solo images.

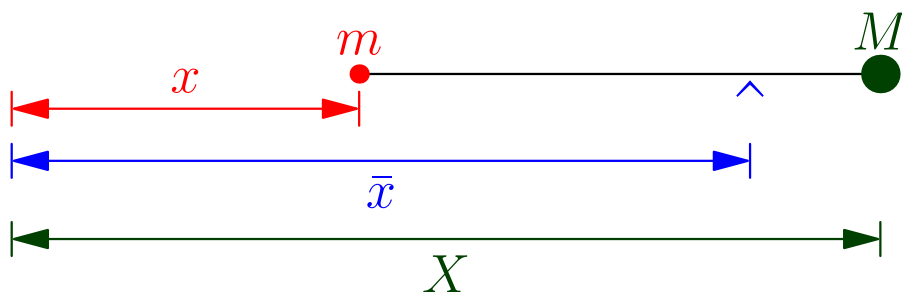
A `tikz` image recycled from above, now 40% width, with 40% left margin, 20% right margin.



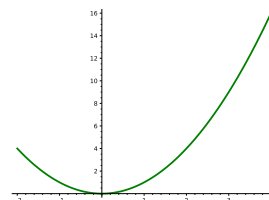
A `pgfplot` image recycled from above, now 20% width, with 40% left margin, 40% right margin, and no longer legible.



An Asymptote image, with zero layout control, so 100% width.



A Sage example, pushed to the right margin.



10.12 Technical Details

The table below is a summary of how graphics and images are specified, constructed and manipulated. Additional processing is indicated by reference to the Python script `pretext`. Images need to be placed relative to the $\text{\LaTeX}$ file that includes them during compilation, and placed relative to the HTML files which reference/include them. Author-provided image files may be placed in any subdirectory, and the `@source` attribute should include the complete relative path with the subdirectory. Files generated by the `pretext` script will be specified in the output using the relative directory `images`, which can be changed. There is no reason author-provided files cannot also be placed in this same directory (presuming no duplicate names). [This table is presently more readable in HTML, the PDF version will improve.]

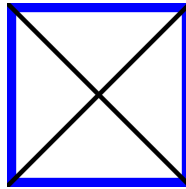
Element	Specification	$\text{\LaTeX}$ /Print	HTML	Notes
<code>image/@source</code>	full relative path, w/ extension	directly included	directly included	author
<code>image/@source</code>	full relative path, w/o extension	presumes PDF	presumes SVG	author
<code>image/latex-image-code</code>	$\text{\LaTeX}$ -compatible source	directly included	SVG via <code>pretext</code>	e.g. tikz
<code>image/sageplot</code>	Sage code	PDF via <code>pretext</code>	SVG via <code>pretext</code>	PNG for
<code>image/asymptote</code>	Asymptote code	PDF via <code>pretext</code>	SVG via <code>pretext</code>	

In the early stages of a writing project, it may be best not to track provisional image files built with `pretext` under version control, and just regenerate them periodically (see the `-r` option for `pretext`). As a project matures, then it makes sense to put stable files under version control for collaborators and others. In every case, managing graphics files (and other aspects of production), is much more pleasurable with a script (shell, Makefile, etc.)

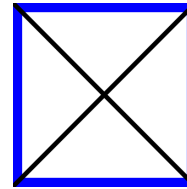
10.13 Accessibility

An `<image>` should either have a non-empty `<description>`, a non-empty `<shortdescription>`, or set `@decorative` to the value `yes`. Some of the following images comply, and some do not. There's not really anything to see here visually, this is testing notifications made elsewhere.

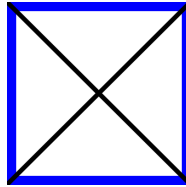
No descrip-
tion, no short-
description, no
@decorative
set to yes



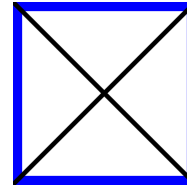
No descrip-
tion, no short-
description,
@decorative
set to yes



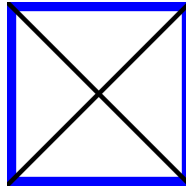
No descrip-
tion, empty
shortdescrip-
tion



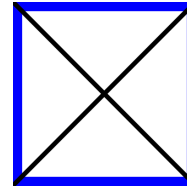
No descrip-
tion, too long
shortdescrip-
tion



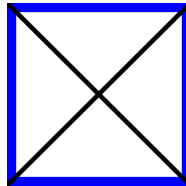
No descrip-
tion, good
shortdescrip-
tion



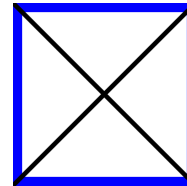
Description,
no shortdescrip-
tion



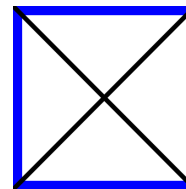
Description,
empty shortde-
scription



Description,
too long short-
description



Description,
good shortde-
scription



11 Further Reading

11.1 Specialized Subdivisions

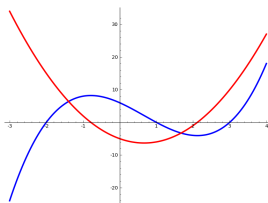
In a longer work you might wish to have some references on a per-chapter basis, or similar. You can make a “references” subdivision anywhere to hold bibliographic items, and you can reference the items like any other item. For example, we can cite the article below [11.4.3, Chapter R], included an indication that a specific chapter may be relevant.

11.2 Exercises

1. No problem here, but the next two are in an “exercise group” with an introduction and a conclusion, along with an optional title. The two problems of the exercise group should be indented some to indicate the grouping.

N.B. An `<exercisegroup>` is meant to hold a collection of (short) exercises with common, shared, instructions. Do not use this structure to subdivide an `<exercises>` division, as you will eventually be disappointed. Instead, use the available, but under development as of 2019-11-02, `<subexercises>`, which *requires* a `<title>`.

Two Derivative Problems. In the next two problems compute the indicated derivative.



You could “connect” the image above with the exercises following as part of this introduction for the exercisegroup.

2. $f(x) = x^3$, $\frac{df}{dx}$. This sentence is just a bunch of gibberish to check where the second line of the problem begins relative to the first line.

We cross-reference the next problem in this exercise group. For the phrase-global form, the common element of the cross-reference and the target should be the exercises division, and not the enclosing exercisegroup: [Exercise 3 of Exercises ??](#).

3. $y = \cos(x)$, y' .

Note that the previous two problems used very different notation for the function and the resulting derivative.

4. This isn’t really an exercise, but an explanation that the next <exercisegroup> has a title and no <introduction>, which once resulted in some aberrant formatting in L^AT_EX output.

Two More Derivative Problems. Some common instructions would go here in the <introduction>

5. $f(x) = x^3$, $\frac{df}{dx}$. This sentence is just a bunch of gibberish to check where the second line of the problem begins relative to the first line.

We cross-reference the next problem in this exercise group. For the phrase-global form, the common element of the cross-reference and the target should be the exercises division, and not the enclosing exercisegroup: [Exercise 3 of Exercises ??](#).

6. $y = \cos(x)$, y' .
7. Compute $\int 3x^2 dx$.
8. One of the few things you can place inside of mathematics is a “fill-in” blank. We demonstrate a few scenarios here. See details on syntax in Subsection ??—the use is identical within mathematics.

- Inside inline math (short, space for x): $\sin(\text{ })$
- Inside inline math (default, space for XXX): $\sin(\text{ })$
- Inside exponents and subscripts (each is space for the string “12”). In this case, be sure to wrap your exponents and subscripts in braces, as would be good L^AT_EX practice anyway: $x^{5+\text{ }}y_{\text{ }}$
- Inside inline math (too long for this line probably, 40 characters long): $\tan(\text{ })$
- So use inside a displayed equation

$$16 \log \text{ }$$

like this one.

- Inside the second line of a multi-line display:

$$y = x^7 x^8$$

$$= x^{}$$

- This fillin has the historical `@characters` attribute for a fillin inside math: $1 + + 4 = 10$, which may be more convenient, but may not side properly in places like subscripts, superscripts, fractions, limits of integrals, and so on.

11.3 More Exercises

This exercises section should start on a new page in PDF output if the standard publisher file is used.

1. This is not a real exercise, we just want to explain that this is another subsection of exercises, which has two consecutive exercise groups.

Exercise Group. Introduction to first exercise group.

2. Only exercise of first group.

Conclusion to first exercise group.

Exercise Group. Introduction to second exercise group.

3. First exercise of second group.
4. Second exercise of second group.

Conclusion to second exercise group.

Exercise Group. An `<exercisegroup>` can have a `cols` attribute taking a value from 2–6. Exercises will progress by row, in so many columns. On a small screen, the HTML exercises may reorganize into fewer columns.

6. $3 + 4 + 5$

Hint. Addition is associative.

5. $1 + 2$

Answer. 12

Solution. First, add 3 and 4 to get 7, then add 5 to arrive at 12.

8. Add seven to eight.

Answer. 15

11. $3 + 4 + 5$

Hint. Addition is associative.

Answer. 12

Solution. First, add 3 and 4 to get 7, then add 5 to arrive at 12.

Proof. A simple argument. ■

And a bit more.

9. $9 + 10$

10. $1 + 2$

12. $5 + 6$

13. Add seven to eight.

Answer. 15

14. $9 + 10$

Answer. 15

15. $1 + 2$

Answer. 12
16. $3 + 4 + 5$

Hint. Addition is associative.

Answer. 12

Solution. First, add 3 and 4 to get 7, then add 5 to arrive at 12.

17. $5 + 6$

Answer. 15

18. Add seven to eight.

Answer. 15

19. $9 + 10$

Answer. 15

20. $1 + 2$

Answer. 15
21. $3 + 4 + 5$

Hint. Addition is associative.

Answer. 12

Solution. First, add 3 and 4 to get 7, then add 5 to arrive at 12.

22. $5 + 6$

Answer. 15

23. Add seven to eight.

Answer. 15

24. $9 + 10$

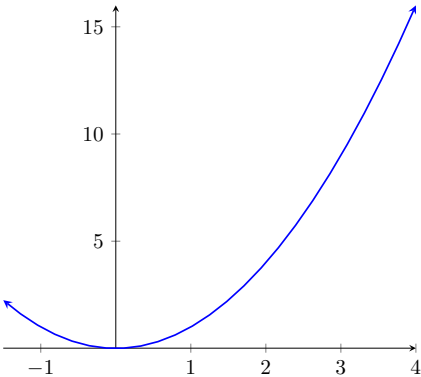
Answer. 15

This feature was designed with short “drill” exercises in mind. With long exercises, or exercises with long hints, answers, or solutions, there is a risk that the $\text{\LaTeX}$ output will have bad page breaks in the vicinity (just before) such an exercise that occupies too much vertical space. Edit, rearrange, or use fewer columns to see if the situation improves.

25. Make a table and a graph for the function $f(x) = x^2$.

Answer.

x	$f(x)$
0	0
1	1
2	4
3	9



11.4 References

These items are here to test basic formatting of references.

- [1] Gilbert Strang, *The Fundamental Theorem of Linear Algebra*, The American Mathematical Monthly November 1993, **100** no.9, 848–855.
- [2] J. B. Conrey and D. W. Farmer, *Mean values of L-functions and symmetry*, Internat. Math. Res. Notices no.17 (2000) pp.883-908.
- [3] Robert A. Beezer, *A First Course in Linear Algebra*, 3rd Edition, Congruent Press, 2012.
An online, open-source¹ offering.
- [4] H. Davenport, *Multiplicative Number Theory*, GTM, **74** Springer-Verlag New York, NY, (2000) xiv+177, A note may accompany a bibliographic item, such as saying the manuscript is under review. But it cannot contain any formatting..
- [5] Alexander Rosswell, *Diffeomorphisms of Penciled Fiber Bundles*, Mathematicians of America (2020), **2** no.6, 884–888.
- [6] Ibid., *Diffeomorphisms of Penciled Fiber Bundles, Part 2*, Mathematicians of America (2021), **3** no.4, 102–103.

This is a conclusion, which has not been used very much in this sample. Did you see that the entry for [11.4.3] has a short annotation? So you can make annotated bibliographies easily.

12 List Calisthenics

12.1 Lists, Generally

Use `ol` to make an ordered list, and `ul` to make an unordered (bulleted) list. In both cases, use `li` for each entry. If an entry contains more than one paragraph, then each must be wrapped in `p`.

This section contains nested lists, to demonstrate how they get assigned labels (numbering, symbols). But we begin with two simple lists, demonstrating an ordered list and an unordered list. See the end of section for an example of a description list. Note in the source the optional use of a paragraph (`p`) for the list items of the list of colors.

1. First.
2. Second¹.
3. Third.
 - Red²
 - Green
 - Yellow
 - Purple

Next, we have a list with no customization and multiple levels to test the defaults. $\text{\LaTeX}$ allows a maximum of four levels of ordered/numbered lists, and a total of six levels if some unordered lists are mixed in. Note that to have nested lists you *must* structure your list items as paragraphs, since a list may only appear within a `<p>` element.

¹A gratuitous footnote to test prior bug confusing this with a REMARK-LIKE `<note>`.

¹Footnote in an unstructured list item

²One of our favorite colors

1. *A title on a top-level item.*
Level 1, first.
2. Level 1, second.
 - (a) Level 2, first.
 - (b) Level 2, second.
 - i. Level 3, first.
 - ii. Level 3, second.
 - A. Level 4, first.
 - B. Level 4, second.
 - C. *Title on xref'ed list item.*
Level 4, third.
 - iii. Level 3, third.
 - (c) *A title on a nested item.*
Level 2, third.
3. Level 1, third.

Items in ordered lists (only) may be given an `xml:id` and then may be the target of an `xref`. We test three here, referencing down into the hierarchy above. Level 1, second: [2](#). Level 3, second: [2.b.ii](#). Level 4, third: [2.b.ii.C](#). Note that if a list item of an ordered list is contained within a list item of an unordered list, then its number will not be defined.

And now a four-level deep unordered list with the default labels supplied by PreTeXt (disc, circle, square, disc). Again, the default order for Markdown/Jupyter (disc, square, circle, circle) is different than for L^AT_EX and HTML (disc, circle, square, disc)

- *A title on a top-level item.*
Level 1, first.
- Level 1, second.
 - Level 2, first.
 - Level 2, second.
 - Level 3, first.
 - Level 3, second.
 - Level 4, first.
 - Level 4, second.
 - Level 4, third.
 - *A title on a nested item.*
Level 2, third.
 - Level 2, third.
- Level 1, third.

And a total of six levels with a mix of ordered and unordered lists, the most that out-of-the-box-L^AT_EX is able to handle.

1. Level 1, first.
2. Level 1, second.

- (a) Level 2, first.
- (b) Level 2, second.
 - Level 3, first.
 - Level 3, second.
 - i. Level 4, first.
 - ii. Level 4, second.
 - A. Level 5, first.
 - B. Level 5, second.
 - Level 6, first.
 - Level 6, second.
 - Level 6, third.
 - C. Level 5, third.
 - iii. Level 4, third.
 - Level 3, third.
- (c) Level 2, third.

3. Level 1, third.

Now, nested lists with the defaults replaced by custom choices. First, an ordered list, three deep, upper Roman numerals, then upper-case Latin, then more traditional Arabic numerals on the three elements of the third level. Note the adornments of the labels will not currently be rendered by [WebKit](#)-based browsers (such as Safari) when viewing HTML output.

I Level 1, first.

II Level 1, second.

- ++A Level 2, first.
- ++B Level 2, second.
 - 1) Level 3, first.
 - 2) Level 3, second.
 - 3) Level 3, third.
- ++C Level 2, third.

III Level 1, third.

A nested unordered list, with labels given as squares on the outer list and nothing (blank) on the inner lists.

- Level 1, first.
- Level 1, second.
 - Level 2, first.
 - Level 2, second.
- Level 1, third.

A nested ordered list, to test intramural cross-references.

1. Level 1, first.
2. Level 1, second.

Level 2, first.

Level 2, second.

3. Level 1, third. With a cross-reference to second list item, [2](#).

4. Level 1, fourth. Whose number should not change when the knowl just prior is opened.

An ordered list may begin at zero by using a numeral zero in the `@label` attribute, instead of numeral one.

(1), start=0 First

(2), start=0 Second

-1-, start=0 Uno

-2-, start=0 Dos

-3-, start=0 Tres

(3), start=0 Third

The next definition is very poorly worded. It is meant to test leading off with a list (bad form), for which L^AT_EX normally begins right after the heading.

Definition 12.1 Group.

- a) There is a binary operation, denoted “.”.
- b) The operation is associative.
- c) There is an identity element, e .
- d) For every element b , there is an element c (the inverse), such that

$$b \cdot c = c \cdot b = e.$$

If these conditions are met for a set G , then we say G is a **group**. ◆

Exercises and References are specialized subdivisions you can put anywhere. They are implemented as top-level lists, so should share behavior. For example, an exercise may have many parts and when expressed as a list, should have the expected labels.

Similarly, References may have lists in their annotations. Unlikely? But possible.

The next two subdivisions are an Exercises subdivision and a References subdivision, which have lists within an exercise and a bibliographic item (respectively).

12.2 List Spacing, I

This is a short list that ends a subsection, so can be used to address the necessary spacing. We also test two XML elements separated by a space (which should not go missing).

- (a) One item. words to check that long entries
in a two column list look good.
- (b) *Two ducks*.
- (c) Three items. Plus a few more
- (d) Four items.

- (e) Another long entry that simultaneously tests that long entries look good in a list, and also tests an odd number of entries in a two column list.

12.3 List Spacing, II

This is another short list that ends a subsection, so can be used to address the necessary spacing.

- Uno item.
 - Dos items.
 - Tres item.
 - Quattro items.
- List column test.** An aside in a multi-column list. We don't want this to float away from its context and/or break the layout. This also tests if list title formatting affects things inside the list.

And a paragraph after that list so that spacing can be checked.

12.4 List items containing only inline math

Testing list items containing only math.

There are many places where it makes sense to have a list of mathematical terms, or possibly equations. For example, one might wish to provide a list of derivative formulas. With such lists, the author may wish to have display mathematics, but almost certainly they don't want it centered. One can work around this by using the $\text{\LaTeX}$ `\displaystyle` command. However, it would be nice if a list item containing only math used display mode by default.

1. A list item containing some text in a paragraph, as well as some inline math: $\int_a^b x^2 dx = \frac{x^3}{3}$.
2. A list item with text and math $\int_a^b x^2 dx$, not in a paragraph.
3. $\frac{d}{dx} \tan(x) = \sec^2(x) = \frac{1}{\cos^2(x)}$
4. $\sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}$

Now, a p that isn't in a list, followed by a list that's in a p.

- $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ A list item starting with some math, followed by text, all in a p
- $\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$

The above assemblage had some lists in it, just to see what will happen. While we're at it, we might try adding lists that are in a list.

List 12.2 A list of items, some of which contain math

1. A first list item, containing some text. The next list item will contain only math, with the `m` tag inline with the `li` tag.
2. A list item with text and math $\int_a^b x^2 dx$, not in a paragraph.
3. $\int_a^b f'(x) dx = f(b) - f(a)$
4. The next two list items will contain, respectively, a list item containing only math, where the math is on a new line, then the same again, but with two new lines, and a list item containing math within a `p`, first inline, and then after a line break.
5. $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$
6. $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$
7. $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$
8. $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$

And now, a list in a paragraph.

- A paragraph that begins with text, then some math: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ And now some more text. The next two list items contain:

1. Math only, inline.
2. Math only, with a newline.
3. Math only, but in a paragraph. Also the next item on this list has math, just to see what happens in a nested list.

4. $\int_a^x \frac{d}{dt} f(t) dt = f(x) - f(a)$

- $\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$

- $\tan \theta = \frac{\sin \theta}{\cos \theta}$

- $\frac{a}{b} > \sum_{i=3}^{76} \frac{x^2}{y^2}$

Inclusion of any text other than math will kill the automatic display style. For example, this would happen if one were to add punctuation after the math.

- $\int_a^b \frac{\sin(x)}{x} dx,$

- $\int_a^b \frac{\sin(x)}{x} dx$

List items can have titles. We try that here, along with testing list items structured with paragraphs.

- With `\displaystyle` added automatically.

$$\sum_{n=1}^{\infty} ar^n$$

- Two paragraphs.

$$\sum_{n=1}^{\infty} ar^n$$

$$\sum_{n=1}^{\infty} ar^n$$

- One paragraph, extra text.

So, $\sum_{n=1}^{\infty} ar^n$

- Two elements (only).

$$\text{\LaTeX}\sum_{n=1}^{\infty} ar^n$$

12.5 Difficult List Items

In Subsection ?? we were careful about lone bits of math inside list items. The `<cd>` element is used with indentation, which is likely superfluous inside a list item that is already being indented. Here we test lone `<cd>` elements inside of list items in various configurations.

Unordered list, one-deep.

- Foo Bar Foo
Bar Foo Bar
- This list item is a long paragraph with a `<cd>` in the middle which should be indented some to indicate its participation in the paragraph.

Foo Bar Foo
Bar
Bar Foo Bar
Foo

This list item is a long paragraph with a `<cd>` in the middle which should be indented some to indicate its participation in the paragraph.

- Foo Bar Foo
Bar
Bar Foo Bar
Foo

Intervening paragraph, to illuminate spacing at both the top and bottom of a list. Intervening paragraph, to illuminate spacing at both the top and bottom of a list. Intervening paragraph, to illuminate spacing at both the top and bottom of a list. Intervening paragraph, to illuminate spacing at top and bottom of a list.

Ordered lists, two-deep, mixed.

1. First item, outer level.
 - (a) First inner item, cd only
Foo Bar Foo
Bar
Bar Foo Bar
Foo
 - (b) Second inner item, a paragraph in a list item.

- (c) Third inner item, cd only
 - Foo Bar Foo
 - Foo Bar Foo
 - Bar Foo Bar
- 2. cd inside second item, outer level
 - Bar
 - Bar Foo Bar
 - Foo

12.6 Description Lists

Use `dl` to make a description list. Inside of those tags, use `li` for each entry. Then, use `title` to specify the term being described and `p` to specify the description.

A “description” list has a short term or phrase that is prominent, followed by a short description. It is modeled on the lists of similar structure in both $\text{\LaTeX}$ and HTML. It makes for a nice medium-weight way to define terms, somewhere in-between the `term` tag which just makes a term prominent in a sentence, and a `definition`, which is set off, has a heading, a number, and a title. Do not try to manage the separation between the title and the description by employing punctuation (but you can include a question-mark or exclamation-point if necessary). For example, do not include a colon to the end of the title. This example is from Bob Plantz.

Central Processing Unit (CPU)	Controls most of the activities of the computer, performs the arithmetic and logical operations, and contains a small amount of very fast memory. This is a second paragraph that should appear indented in PDF output.
Memory	Provides storage for the instructions for the CPU and the data they manipulate.
Input/Output (I/O)	Communicates with the outside world and with mass storage devices (e.g., disks).
Bus!	A communication pathway with a protocol specifying exactly how the pathway is used. (The punctuation is just for testing.)
$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$	A geometric series. The formula is valid if $ x < 1$.

Some presentations can be assisted by a hint from the author about the lengths of the titles. You can choose to provide a `width` attribute on a `dl` element with possible values `narrow`, `medium`, and `wide`. The value refers (somewhat confusingly) to the distance between the left margin and the description. The default is `medium`, which is illustrated above. An example with `narrow`:

Red	The color of the sun at sunset.
Blue	The color of a clear sky. Also a synonym for “depressed or sad”, the title of a 1971 Joni Mitchell album (and more than a dozen other musical albums), the period of Picasso’s work between 1901 and 1904, and much more!
Aqua	The color of shallow tropical waters. ³

Math x^2 Sorry, not a color but testing titles with math in them.
“i” before “e” except after “c,” unless it sounds like “a” as in “neighbor” and “weigh”

Get feisty about that weird counterfeit rule: seize the day and don’t have a heifer, man.

Avocado Avocado is the the color with hex code #568203, and also the main ingredient in guacamole.

Magenta Magenta is a color, and a character in Rocky Horror.

Zymurgist

A scientist who studies the chemical process of fermentation in brewing and distilling. Also the alphabetically last 9-letter word in the English language.

Byzantium

Byzantium is the the color with hex code #702963, and also an ancient Greek city which later became known as Constantinople, and today is called Istanbul.

Convection

Circulating motion in a fluid.

Elementary

No literary detective ever said “Elementary my dear Watson.” In particular, Sherlock Holmes never said that.

Understand

Perceive the intended meaning of.

Washington

A state, a district, the man on the US \$1 bill and on the US quarter. Did you ever notice that on the US dime, the value is stated as “one dime”? But how is one to know that a dime is worth 10 cents?

Aquamarine

Aquamarine is a color, and a mineral.

Those who cannot remember the past are condemned to repeat it.

George Santayana wrote those words in 1905. A similar aphorism is misattributed to Winston Churchill. The idea is embodied in the 4th principle: PreTeXt respects the good design practices which have been developed over the past centuries.

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

The Riemann ζ -function is defined by a Dirichlet series, valid for $\Re(s) > 1$.

main() is a void function

A dl with width="narrow" might be a useful way to give commentary on a program listing.

And the same example with wide:

Red The color of the sun at sunset.

³On a sunny day! (Testing footnotes in description lists for L^AT_EX output.)

Blue	The color of a clear sky. Also a synonym for “depressed or sad”, the title of a 1971 Joni Mitchell album (and more than a dozen other musical albums), the period of Picasso’s work between 1901 and 1904, and much more!
Aqua	The color of shallow tropical waters. ⁴
Math x^2	Sorry, not a color but testing titles with math in them.
“i” before “e” except after “c,” unless it sounds like “a” as in “neighbor” and “weigh”	Get feisty about that weird counterfeit rule: seize the day and don’t have a heifer, man.
Avocado	Avocado is the the color with hex code #568203, and also the main ingredient in guacamole.
Magenta	Magenta is a color, and a character in Rocky Horror.
Zymurgist	A scientist who studies the chemical process of fermentation in brewing and distilling. Also the alphabetically last 9-letter word in the English language.
Byzantium	Byzantium is the the color with hex code #702963, and also an ancient Greek city which later became known as Constantinople, and today is called Istanbul.
Convection	Circulating motion in a fluid.
Elementary	No literary detective ever said “Elementary my dear Watson.” In particular, Sherlock Holmes never said that.
Understand	Perceive the intended meaning of.
Washington	A state, a district, the man on the US \$1 bill and on the US quarter. Did you ever notice that on the US dime, the value is stated as “one dime”? But how is one to know that a dime is worth 10 cents?
Aquamarine	Aquamarine is a color, and a mineral.
Those who cannot remember the past are condemned to repeat it.	George Santayana wrote those words in 1905. A similar aphorism is misattributed to Winston Churchill. The idea is embodied in the 4th principle: PreTeXt respects the good design practices which have been developed over the past centuries.
$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$	The Riemann ζ -function is defined by a Dirichlet series, valid for $\Re(s) > 1$.
main() is a void function	A dl with width=“narrow” might be a useful way to give commentary on a program listing.

You can nest description lists, though items might get crowded horizontally.

⁴On a sunny day! (Testing footnotes in description lists for L^AT_EX output.)

We expand our computer list a bit.

Expanded CPU	Controls most of the activities of the computer, performs the arithmetic and logical operations, and contains a small amount of very fast memory.
Arithmetic	The kind of computations that give a computer the name “computer.”
Logic	Operations that make decisions, “if this, then that”, plus: and, or, and not.
Memory	Space to store information and results, permanently or temporarily.
This is a second paragraph that should appear indented in PDF output.	
Memory	Provides storage for the instructions for the CPU and the data they manipulate.
Input/Output (I/O)	Communicates with the outside world and with mass storage devices (e.g., disks).
Bus!	A communication pathway with a protocol specifying exactly how the pathway is used. (The punctuation is just for testing.)
$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$	A geometric series. The formula is valid if $ x < 1$.

12.7 Named Lists

A list can be wrapped with a <list> element, so that it earns a number, can be given a title and have an introduction and conclusion. Cross-references to individual list items get a bit involved as they are prefixed with the number of the list and then the number of the item, so conceivably you could get a number like 4.5.3:2.a.ii. The colon is used to indicate the transition from the number of the list within divisions and the numbers coming from the list hierarchy, since it has two small dots.

List 12.3 Colors of the Rainbow

Because the colors are always in the same order, an ordered list is natural here. The colors change continuously, but are often divided up into large ranges that human perception can easily distinguish.

1. Red
2. Orange
3. Yellow
4. Green
5. Blue
6. Indigo

7. Violet

So some people use the acronym ROY-G-BIV to remember this sequence.

This next list is used for testing cross-references to it. See Section ??.

List 12.4 A named list of targets

This is the introduction to this named list, which references an item within, via the hybrid `@text` attribute: [Item B.c](#). At one time this paragraph was inadvertently centered—that bug has been fixed.

- A
 - i A and i
 - ii A and ii
 - iii A and iii
- B
 - a B and a
 - b B and b
 - c B and c (target of some cross-references)
- C
 - The next three cross-references point to a list item, just above. It is interesting because the list is named, hence numbered. The global reference uses the full number, while the local reference uses the number from within the list. The hybrid reference recognizes that the target is within the same named list, so the number can be shorter. An identical hybrid cross-reference appears within the `<introduction>` to this list, an immediately following, but outside the `<list>`.
 - Cross-reference within named list (global): [Item 12.4:B.c](#)
 - Cross-reference within named list (hybrid): [Item B.c](#)
 - Cross-reference within named list (local): [Item B.c](#)
 - - 1 C and bullet and 1
 - 2 C and bullet and 2
 - 3 C and bullet and 3
 - C and bullet
 - C and bullet

This is a paragraph just outside the preceding named list, which references an item within, via the hybrid `@text` attribute: [Item 12.4:B.c](#).

This is a paragraph with three lists contained within it. For HTML output we have to “inside-out” the lists.

1. A one item ordered list.

In other words, the text before, after, and between, needs to each be encapsulated as an HTML `p` element of its own.

- A one item unordered list.

Including definition lists.

Define Me A one item definition list.

That’s all!

A one item list, whose item is a paragraph with two contained ordered lists, separated by text.

- Introductory text.

A First item, first list.

Intermediate text.

a First item, second list.

Concluding text.

12.8 Testing List Decompositions

A list in a paragraph is a construction in HTML that browsers try to correct, which leads to unpredictable results, so we have to decompose an author's paragraph with lists into a sequence of HTML paragraphs, interrupted by lists. This subsection is only relevant to HTML output, and only for testing.

1. This paragraph opens with an ordered list.
2. Testing the id, and other info that should be at the top of the paragraph.

Now the paragraph continues, and we have an index item here, so we can test cross-references back here.

12.9 List Column Testing

This is a list arranged into two columns with some intentional layout challenges. The math is too long to fit in one column and can't be wrapped - it reduces the number of columns in its row. The long text items can be wrapped and stay within their column.

- | | |
|---|---|
| (a) One item. | (e) A long item that can be wrapped over multiple lines. |
| (b) <i>Two ducks.</i> | (f) Four items. |
| (c) | (g) |
| $\begin{array}{rcl} 3u + 1v + 3w + 2x + y & + 3z & = 10 \\ 2u + 2v - 2w + x & + z & = 6 \\ 1u - 3v + 1w + x & + 3y + 2z & = 13 \end{array}$ | (h) Another long entry that simultaneously tests that long entries look good in a list, and also tests an odd number of entries in a two column list. |
| (d) Short item. | |

12.10 Exercises (with lists)

1. This exercise should have several parts, and labels should follow the defaults for second-level lists (since the exercise is numbered according to the top-level default).
 - (a) Exercise 1, first part.
 - (b) Exercise 1, second part.
 - i. Exercise 1, second part, first refinement.
 - (c) Exercise 1, third part.

2.

Table 12.5 Table Alignment Example

1111, 2222		3333
aaaa	bbbb,cccc	
AAAA	BBBB	CCCC

This exercise (a list item really) has a `table` first. Default $\text{\LaTeX}$ aligns it vertically above the exercise number. Placement here tests correcting that alignment.

A small test of cross-references to subsidiary parts of exercises. Exercise 1, third part: [12.10.1.c](#). Exercise 1, second part, first refinement: [12.10.1.b.i](#).

12.11 References (with lists in Annotations)

- [1] Some book would be listed here.
 Here is the annotation and an ordered list as part of that annotation.
- (a)Book 1, first part.
 - (b)Book 1, second part.
 - (c)Book 1, third part.

13 Table Calisthenics

c = 832
c

That was a Sage cell just now, which has nothing to do with tables. But we needed someplace to test placement right after a division heading. Carry on.

A very minimal table, hence with left-justified cells, no borders. We do wrap the tabular element in a table element to get centering, numbering and a caption. Footnotes inside cells are tested here.

Table 13.1 Some Colors

Red	Green ¹	Yellow
Blue	White	Pink

Note that tables may be constructed using the [\$\text{\LaTeX}\$ Complex Table Editor](#) tool online at [latex-tables.com](#) and then exported in PreTeXt syntax.

Tables can be used and abused many ways. We describe long division of polynomials by using vertical and horizontal borders on individual entries of a `<tabular>`. The division lines are slightly thicker than the subtraction lines. This is a good example of the typical abuse of tables for horizontal and vertical layout. At least we have called it a “Figure,” not a “Table”.

¹Green can be a very sick looking color.

$$\begin{array}{r}
 x \quad + \quad 2 \quad \Bigg| \quad \begin{array}{r} x^2 \quad - \quad 3x \quad - \quad 8 \\ x^2 \quad + \quad 2x \end{array} \\
 \hline
 \begin{array}{r} -5x \quad - \quad 8 \\ -5x \quad - \quad 10 \end{array} \\
 \hline
 2
 \end{array}$$

Figure 13.2 Polynomial Long Division

An example of aligning table cells' contents horizontally. See the source for comments.

Table 13.3 Horizontal Alignment Example

1234567890	1234567890	1234567890	1234567890
[First	Second	Third	Fourth
A	B	C	D
1	2	3	4

Example from above, but now with horizontal rules, plus an extra row to test the bottom border. See the source for comments.

Table 13.4 Horizontal Rules Example

1234567890	1234567890	1234567890	1234567890
First	Second	Third	Fourth
A	B	C	D
1	2	3	4
1	2	3	4

For a table without a caption, create a `<tabular>` and place it directly within the current division. This will allow control over the horizontal placement, but without a caption, there is no number, and the tabular cannot be cross-referenced.

One

Same example as before, but now with vertical rules. See the source for comments.

Table 13.5 Vertical Rules Example

1234567890	1234567890	1234567890	1234567890
First	Second	Third	Fourth
A	B	C	D
1	2	3	4
1	2	3	4

Table 13.6 Progressively Thicker Rules Example

1111	2222	3333
aaaa	bbbb	cccc
AAAA	BBBB	CCCC

Table 13.7 Column Span Example

1111, 2222		3333
aaaa	bbbb,cccc	
AAAA	BBBB	CCCC

A list whose first item is a table. In $\text{\LaTeX}$ output a `\leavevmode` is necessary to keep this organized (item number, *then* table as content).

1. **Table 13.8 Table Alignment Example**

1111, 2222		3333
aaaa	bbbb,cccc	
AAAA	BBBB	CCCC

Example 13.9 Example Environment with Leading Table.

Table 13.10 Column Spans, No `col` Elements, Nine Columns

1	2+3	4	5+6+7	8+9				
1	2	3	4	5	6	7+8	9	
1	2	3	4	5	6	7	8	9

This example tests several things. In $\text{\LaTeX}$ output, figures, tables, listings and side-by-sides are “floats” whose placement can migrate, but we have tries to supress this behavior. However, a float that is the first item of an “environment” (like a theorem or an example) can still float to a position *before* its title. If that does not happen here, then our additional defenses are working.

This example also checks that the total number of columns is correctly computed from the first row, which features several `colspan` attributes. ▲

A bare minimum table (one row with one cell) to test edge cases:

Table 13.11 One entry table

One

Table cells with a fixed width where text wraps are known as “paragraph cells”. A cell will be created as a paragraph cell if and only if it has `<p>` children. And such cells should *only* have `<p>` children. The width of a paragraph cell is determined by a `width` attribute on the corresponding `<col>` (as a percentage). If the column has a non-paragraph cell with contents that are wider than the paragraph cells, results will be undesirable. There is presently no implementation for a paragraph cell that has a `colspan` greater than 1, although cells with `colspan` greater than 1 that are above or below a paragraph cell will behave. Setting `width` on a `<col>` that has no paragraph cells may produce unexpected results. A `valign` for the parent `<row>` (or the ambient `<tabular>`) can control vertical alignment (top, middle, or bottom). A paragraph cell’s `halign` attribute (left, center, right, or justify) controls how the text is justified. Cells inherit `halign` from `<row>`, `<col>`, and `<tabular>` in that order of preference. In a non-paragraph cell where `halign='justify'`, the horizontal alignment will match the behavior of `halign='left'`.

Table 13.12 Time Units

Unit	Stands For	Definition	Roughly
s	second	the duration of 9192631770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom an extraneous paragraph just to demonstrate the inter-paragraph formatting.	the time it takes you to say the phrase “differential calculus”
min	minute	exactly 60 seconds	how long it takes to microwave a full dinner plate from the refrigerator
h	hour	exactly 3600 seconds; exactly 60 minutes	the length of one episode of a premium cable television show

Table cells can have multiline content using `<line>` elements. This is not the same thing as a paragraph cell—line breaking will happen precisely where the author tells it to. A `<line>` will not break, even on a narrow screen. If a cell uses a `<line>`, it must only use a sequence of `<line>`s and no other content. As with paragraph cells, you can use a `valign` attribute for the row.

Table 13.13 Dr. Seuss lines

One Fish		Look at me!
Two Fish		Look at me!
Red Fish	I am the Lorax.	Look at me NOW!
Blue Fish	I speak for the trees.	It is fun to have fun.
	Self-referential: Table ??	But you have to know how.

This is a table torture test with many combinations of `halign`, `valign`, `colspan`, `<p>` children, and `<line>` children.

Table 13.14 Table Torture Test

							Cell too wide	
Lf md	Lef mid par cel	Rt md	Rig mid par cel	Cn md	Cen mid par cel	Js md	Jus mid par cel jus mid par cel	
Colspan=2 lef mid with lines		Colspan=3 rig mid			Lines Between Par	Lines Between No Par	Par in row with lines	
L t	Lef top par cel	R t	Rig top par cel	C t	Cen top par cel	J t	Jus top par cel jus top par cel	
L b	Lef bot par cel	R b	Rig bot par cel	C b	Cen bot par cel	J b	Jus bot par cel jus bot par cel	
Colspan=3 lef bot			Colspan=2 rig bot with lines		Lines Under Par	Lines Under No Par	Par in row with lines	

And now a <sidebyside> with a <table> and a <tabular> to check that width is scaled appropriately. See Section ?? to learn about <sidebyside>s.

A1.S1	All legislative Powers herein granted shall be vested in a Congress of the United States, which shall consist of a Senate and House of Representatives. Should be 50% of 45% except perhaps on small screens.
A1.S2.C1	The House of Representatives shall be composed of Members chosen every second Year by the People of the several States, and the Electors in each State shall have the Qualifications requisite for Electors of the most numerous Branch of the State Legislature. Should be 50% of 55% except perhaps on small screens.

Figure 13.15 Some text from the US Constitution

Tables are formed in L^AT_EX output with copious use of the \multicolumn macro to override more global alignment settings, and to spread the content of

one cell across several columns. However, sometimes $\text{\LaTeX}$'s special characters have behaved badly in this situation. So the table below, two items per row, is just designed for $\text{\LaTeX}$ testing. But of course, it should still render fine in other formats. The three test cases are from ??, but without 50 alphabetic characters and 8 digits, which should not be problems in this context. That section says more about making a comprehensive, yet legal, test URL. In order to test the use of a percent sign (%) in a URL, we follow it by two hex digits, specifically, 58, which is a way to represent the character X in a URL. The first column's entries are *forced* to be wrapped in a $\text{\multicolumn}$ by specifying their horizontal alignment. The second column's entries *will not* be wrapped in a $\text{\multicolumn}$. So the two columns will look identical, other than the first having a left alignment, and the second has the default center alignment. (This table is known to render poorly in a Jupyter notebook. The cause is four dollar signs present in rows 1 and 3, and is explained in Subsection ??.)

Table 13.16 Problematic Table Cells for $\text{\LaTeX}$

1	$\text{\multicolumn}{1}{l}{09az\%-._~:/%5B\%5D@!\$&'()*+,\;=#?}$	$09az\%-._~:/%5B\%5D@!\$&'()*+,\;=#?$
2	$\text{\multicolumn}{1}{l}{e.com/09az\%58-._~:/%5B\%5D@!\$&'()*+,\;=#?}$	$e.com/09az\%58-._~:/%5B\%5D@!\$&'()*+,\;=#?$
3	$\text{\multicolumn}{1}{l}{example.com/09az\%58-._~:/%5B\%5D@!\$&'()*+,\;=#?}$	$example.com/09az\%58-._~:/%5B\%5D@!\$&'()*+,\;=#?$

Now, the same table repeatedly, but with different headers. No care has been taken with alignment or rules, which could improve how these look.

Table 13.17 No Headers

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.18 One Row Header

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.19 One Row Header, Multiline

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.20 Two Row Headers

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.21 One Vertical Row Header

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.22 One Vertical Row Header, Multiline

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.23 Two Vertical Row Headers

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.24 Test of vertical header orientation and padding with borders

	0 (<i>MTH111</i>)	1 (<i>MTH112</i>)	2 (<i>CS160</i>)	3 (<i>CS161</i>)	4 (<i>CS162</i>)	5 (<i>CS205</i>)
0 (<i>MTH111</i>)	0	1	0	1	0	0

Table 13.25 One Row Header, with Rules

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

Table 13.26 One Row Header, Multiline, with Rules

State	Population	Area (sq. mi.)	Statehood (Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

The next table has a progression of thicker rules in the header, plus a progression of thicker rules across the columns. For testing, not for aesthetics.

Table 13.27 Two Row Header, Many Rules

State	Population	Area	Statehood
		(sq. mi.)	(Year)
Washington	7,614,893	71,362	1889
Oregon	4,217,737	98,381	1859
California	39,512,223	163,696	1850

We now finish this section with some *long* tables. Ones that will not fit on a single printed page. So this is only of interest when producing this sample article as a PDF. First a “naked” tabular, which should force a new page to start, and then still overrun the end of the page.

[illegible]

Now the same `<tabular>`, but within a `<table>`. Behavior should be similar, a new page and then it overruns the bottom of the page.

Table 13.28 A Lot of Colors

[illegible]

When you wish to allow a `<tabular>` to split itself at a page break, you can add the attribute `@break` with the value `yes`. Certainly this will control a `<table>` or `<tabular>` which is longer than a page, but will also allow a shorter one to break. This might be useful in a draft stage before undertaking page-fitting.

Here is the long <tabular> again, but with the @break attribute set to yes.

[illegible]

Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink

Here is the long `<table>` again, but with the `@break` attribute set to `yes` on the `@tabular`.

Table 13.29 A Lot of Colors, Breaking

Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink

Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
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Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink
Red	Green	Yellow
Blue	White	Pink

This device is not ideal, as some features of tables are not behaving as expected. More precisely, we switch from the standard L^AT_EX `tabular` environment to the `longtable` environment from the package of the same name when table-breaking is requested. So there may be some undesirable interaction with other packages. For one, full-width horizontal rules seem to become as wide as the page (rather than as wide as the `tabular`). The following table is Table ?? repeated but as a breakable `<tabular>`. The `longtable` package documentation suggests it accomodates the `array` package, but it also seems to make a variety of redefinitions. Furthermore, a panel of a side-by-side cannot be a breakable `tabular`, or a L^AT_EX compilation error occurs.

Table 13.30 Horizontal Rules Example

1234567890	1234567890	1234567890	1234567890
First	Second	Third	Fourth
A	B	C	D
1	2	3	4
1	2	3	4

Here is a consecutive pair of “bare” `<tabular>` to test vertical space between them.

t	2/3	2/303	2/30003	2/3000003
$g(t)$	−1	−1	−1	−1
t	2/5	2/505	2/50005	2/5000005
$g(t)$	1	1	1	1

The `longtable` package allows for headers and footers indicating continued tables. A possible enhancement is to support this feature in the case of a long table.

14 Interactive Elements, Authored in Javascript

When outputting Web page versions, it is possible to embed a variety of dynamic interactive elements. In a L^AT_EX/PDF version, these will necessarily need to be replaced by some static substitute, such as a screenshot. See Section ?? for the specifics of embedding instances of the Sage Cell Server, which is more elaborate, and not entirely similar.

Interactives in this section are those for which you provide code you have authored. Generally, the libraries involved to support this have open licenses, though the player for GeoGebra may be an exception. Creating these assumes some familiarity with HTML and Javascript. See Section ?? for more interactives that are perhaps simpler to create or use.

(2018-06-22) Almost everything in this section is under active development and not stable yet. Feel free to experiment and make suggestions and requests. This page takes a while to completely load, so be patient.

14.1 HTML5 Canvas

HTML5 introduced the `<canvas>` element, which can be thought of a blank slate, a place to draw or write on. So PreTeXt has the `<slate>` element for a similar purpose. Generally, but not exclusively, HTML5 writes on a `<canvas>` using the Javascript language. We demonstrate this approach to interactive diagrams in this subsection.

The following examples are from David Austin's excellent [Understanding Linear Algebra](#) textbook, which can be found at

understandinglinearalgebra.org

David's contribution of examples, and assistance designing the PreTeXt elements is greatly appreciated. Alright, let's learn some linear algebra. Yes, there are some learning opportunities in this subsection.

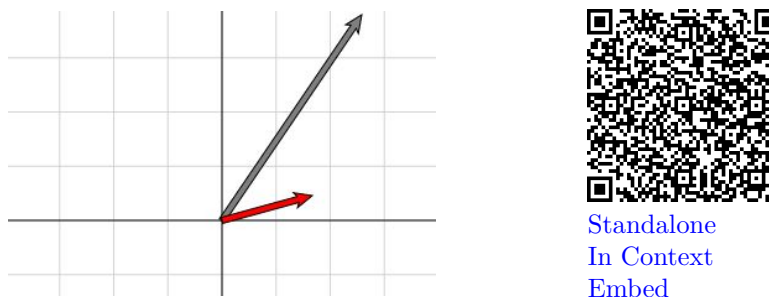
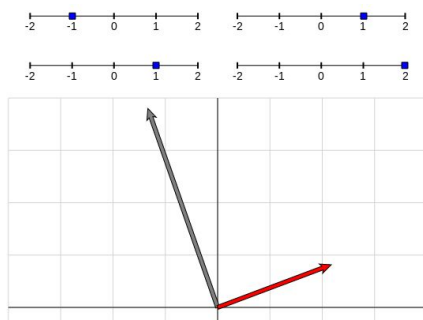


Figure 14.1 A simple eigenvector demonstration

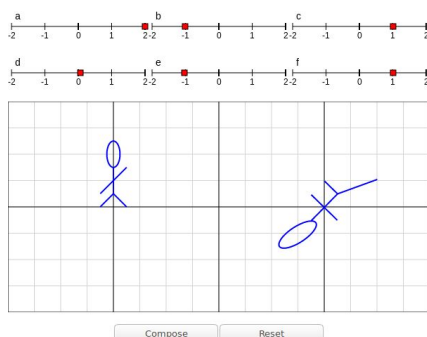
Checkpoint 14.2 The interactive in Figure ?? shows a vector $\vec{x}$ in red, and the matrix-vector product $A\vec{x}$ in grey, for a particular 2×2 matrix A . The four entries of the matrix A are coded into the interactive. Can you deduce A simply by using the interactive? Which theorem is the key?



[Standalone](#)
[In Context](#)
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Figure 14.3 Eigenvector demonstration

The next example has ten `<slate>` elements communicating with each other, and arranged with the layout features of a `<sidebyside>` (see Section ??).



[Standalone](#)
[In Context](#)
[Embed](#)

Figure 14.4 Affine Transformations

Warning 14.5 If your `<interactive>` employs a `<slate>` with a `@surface` attribute whose value is `html`, then you are advised to augment each top-level (HTML) element within the `<slate>` with the attribute:

```
xmlns="http://www.w3.org/1999/xhtml"
```

This will identify *all* of the elements within the `<slate>` as HTML elements and not as PreTeXt elements. The danger is that elements with the same name in both languages, such as `<li>` and `<table>`, will be mis-identified. This could be harmless, but could also create chaos, such as disrupting numbering of PreTeXt elements.

See the source code of this document for examples: Figure ?? Figure ??, Figure ??.

Note that the HTML that is output can vary slightly from your source in small, harmless ways, such as empty (self-closing) elements being output with both an opening and a closing tag. Please report any significant discrepancies. Soon this requirement will be enforced in the code.

It is also possible to add `<script>` elements within an interactive that contain properly escaped JS code. These elements will be placed at the end of the document created to hold the interactive content and can interactive with the other elements within the interactive but can not directly interact with the surrounding page.

Authors are strongly discouraged from trying to incorporate complex code in the form of a `<script>`, but it can be a useful tool to call more complex code that is linked via `@source` on the `<interactive>`.

This example uses a `<script>` to draw “Hello World” to a `<slate>`



[Standalone](#)
[In Context](#)
[Embed](#)

Figure 14.6 A simple embedded script example

This next interactive uses ECMAScript module based code. That will not function unless the `<script>`'s `@type` is set to `module` in the HTML output. So let's set it here and pass it through. It also generates its own HTML elements, so we hopefully don't get a default empty div.



[Standalone](#)
[In Context](#)
[Embed](#)

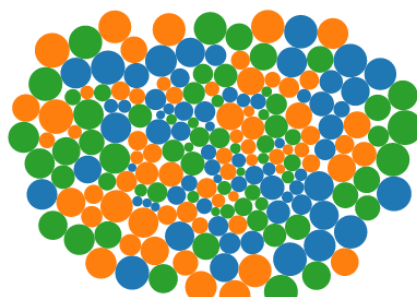
Figure 14.7 A simple JS module test

14.2 D3.js

D3 is a Javascript library for “Data-Driven Documents”, which might greatly enhance some data you wish to display. In short, it uses the animation capabilities of SVG. Available examples seem sensitive to the version of the library, so we have examples using different versions. Use the `@version` attribute on `<interactive>` to specify the version number. The default is 5.

The first example uses the force layout and collision detection from Version 3. (The necessary commands are very different in Version 4.) Pretend you are a working shepherding dog. Can you separate, and catch, one of the herd?

This is adapted from a [block](#) by [Mike Bostock](#) with a GPL license. A similar demonstration, only using an HTML5 canvas is at bl.ocks.org/mbostock/3231307.



[Standalone](#)
[In Context](#)
[Embed](#)

Figure 14.8 Force layout and collision detection

Similar, but different, this demonstration of a graph layout uses Version 4 of the library. Technical notes:

- We have changed the size of the nodes, and their number, to fit in a smaller space.
- The Javascript script uses introspection to size itself, which would be a good general practice. (Though it does not resize itself on window resize, only initial load.) The `<interactive>` also has the `@resize-behavior` attribute set to `responsive`.

This is adapted from a [block](#) by [shimizu](#) with a GPL license.

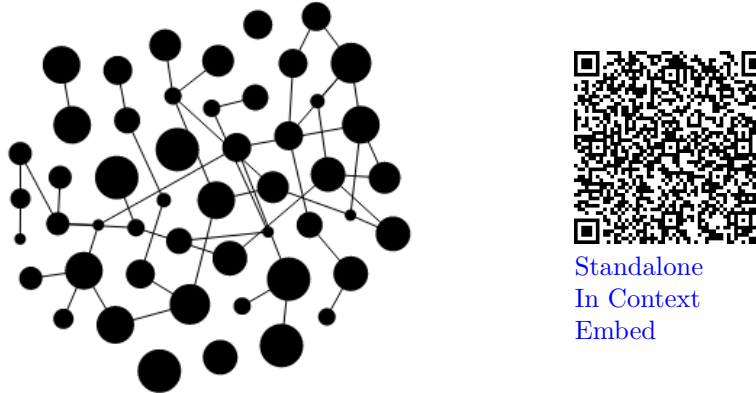


Figure 14.9 Graph Layout

Checkpoint 14.10 Graph Planarity. Can you move the vertices to new locations such that the resulting graph is planar? (In other words, no edges cross?)

Finally an example that actually uses some data. Here is the description from the [original block](#) by [Martin Chorley](#) with an MIT License.

This visualisation uses a D3 force simulation to show the Twitter relationships between the Assembly Members in the Welsh Assembly in terms of the number of times each assembly member has mentioned another assembly member in a tweet.

Twitter relationships were mined on 22/03/2017, and are representative of the conversational relationships on that date. Links between AMs represent a conversational relationship: one AM has mentioned the other. Party colour indicates the direction of the mention.

Hover over the nodes to fade out non-connected nodes.

Rather than using intermediate nodes to create curved links (as in Mike Bostock's block), this adds curves by adding a calculated control point for each edge.

Technical notes:

- Once the nodes organize themselves (automatically in the beginning), they cannot be moved.
- We have adjusted the margins in an attempt to keep names visible on the right side, but without giving up too much space.
- We have adjust the repelling force, and the collision buffer, to better fit the available space.

- This example required its own CSS, which we have included as part of the `<interactive>`.
- The `@aspect` ratio of the `<interactive>` is set to slightly taller than that of the `<slate>` it contains. This accounts for the fact that the contents generated by the js/css in the slate occupy slightly too much vertical real estate.
- The data collected from the Twitter analysis is contained in a JSON file, `mention_network.json`, and where the script loads that file, it has a hardcoded path. So this example is a bit brittle, should that file move.

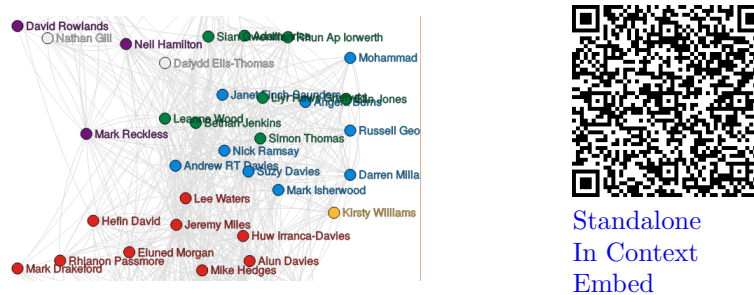


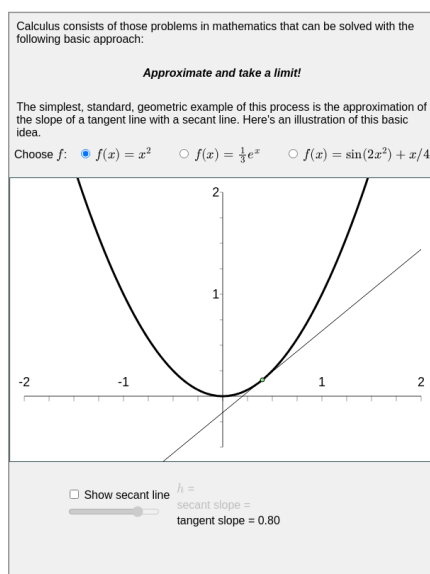
Figure 14.11 Tweet mentions within the Welsh Assembly

14.3 SVG

Entirely similar to using an HTML5 `canvas` element (??), it is possible to control an SVG element with Javascript. This example is from Mark McClure.

Look carefully at the source and rendering of this example as HTML. The functions to choose from via radio buttons, and the change in x , denoted later as h , are being rendered as mathematics by the Javascript MathJax library. However, this cannot be accomplished with simply `$. . . $` nor by simply `\( . . . \)`, but instead by using the `\( . . . \)` and then also wrapping that in a `<span>` element with a `@class` attribute set to `process-math`. (The latter is how we instruct MathJax as to which parts of a page to process, or not). So to have MathJax render a nice x^2 in this context (math inside HTML inside a `<slate>` inside an `<interactive>`) would be accomplished with

```
<span class="process-math">x^2</span>
```



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Figure 14.12 Tangent and secant slopes

Checkpoint 14.13 Changing Secant Lines. When discussing the derivative as a limit, we think of the point of tangency as being fixed (the green point in ??) and the “other” point defining the secant line as changing (the red point in ??). Switch it up! Fix a large value of h (positive or negative) and then change the point of tangency (the green point). Discuss what you observe.

14.4 JSXGraph

[JSXGraph](#) is a “cross-browser JavaScript library for interactive geometry, function plotting, charting, and data visualization in the web browser.” Now a `<slate>` will be what JSXGraph calls a **board**. Again, you use Javascript to write onto a `<slate>`, but have some powerful shortcuts available from the JSX-Graph library. For this reason, PreTeXt calls JSXGraph a “language”, similar in many respects to how Sage is a language, but is really a Python library. So realize that the syntax for using JSXGraph is that of Javascript.

Place Javascript inside a file that is specified with the `@source` attribute of the `<interactive>` element. Then just be certain that `@xml:id` of the `<interactive>` element is passed as the HTML `id` in an (early) call to JSX-Graph’s `initBoard()` method.

The following example was contributed by Rick Roesler. The `<figure>` is comprised of four `<stack>` elements within a `<sidebyside>`. By varying the time in the top box, the reader can observe the displacement, velocity, and acceleration of a ball thrown upward with an initial velocity of 30 m/s.

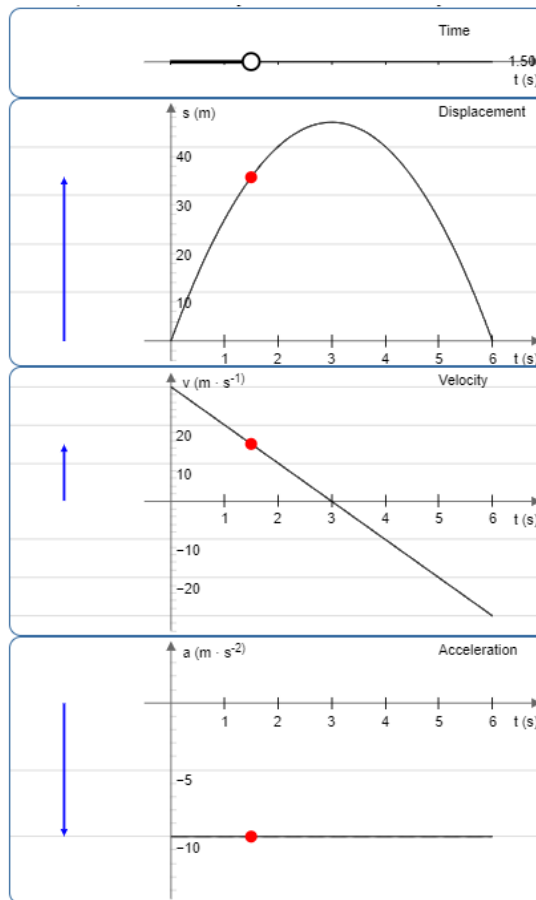
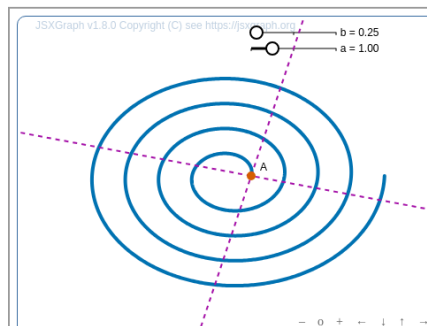


Figure 14.14 1-Dimensional Kinematics

The plot below is the curve $r = a + b\theta$ in polar coordinates, for $0 \leq \theta \leq 8\pi$. It may be manipulated with the sliders to control the shape of the curve. Point A is constrained to the curve, but may be dragged to a new location. At A the tangent line and normal line are plotted as dashed red lines. Use the controls in the lower left to adjust the viewing window. This [Archimedean Spiral](#) is taken from the [JSXGraph example wiki](#). The code could be written in 7 lines. Width is 80% and aspect ratio is 4:3.



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[In Context](#)
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Figure 14.15 The Archimedean Spiral $r = a + b\theta$, $0 \leq \theta \leq 8\pi$

Here is a more elaborate example, from the [JSXGraph Showcase](#), titled [Infinity](#).

There are two active sliders to control the shape and shading of the graphic, and hovering the mouse near one of the edges will highlight the entirety of one